

Surreal Probability and Log-Odds

A multiscale theory of belief, information,
and infinite sampling

Research article prepared for Vladimir Reshetnikov

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Abstract

We develop probability with values in set-sized subfields of the surreal numbers, and logarithmic odds defined by the canonical surreal exponential. The finite theory includes conditioning on infinitesimal events, exact Bayesian updates, a leading-scale calculus with coefficient-sensitive tie breaking, quantitative conditioning stability, soft-max models, relative entropy, proper scoring rules, Gibbs variational principles, and finite-horizon stochastic processes. A conditional real shadow retains information that the unconditional standard part destroys. A finite linear-algebra compression also captures all comparisons between real-valued acts without discarding signed correction coefficients.

Infinite sample spaces require additional structure rather than a change of scalar field alone. We distinguish strong Hahn summation, coefficientwise countable additivity, and regular finitely additive probabilities obtained from ultrafilters or ordered-field compactness. We construct diffuse hierarchical probabilities and posterior kernels, prove two elementary obstructions to coefficientwise independent sampling, and give a regular all-subsets extension theorem after a set-sized field enlargement. An uncountable-support example shows that naive coefficientwise integration can send a pointwise positive surreal-valued function to a negative number. Standard-part and nonstandard/Loeb bridges are treated separately. Throughout, finite algebraic identities, support-controlled series, and probabilistic limit theorems are kept distinct.

Status and provenance. This is an AI-assisted mathematical synthesis with detailed proofs and executable exact finite checks, not an independently refereed or Lean-verified paper. Non-Archimedean probability, lexicographic belief, and extended log-likelihood methods have substantial precedents. No first-in-literature claim or solution of a named open conjecture is made. The repository comparison is pinned to the commit recorded in Section 1. The article was delivered as a standalone file and is now a single-source report of that collection; Section 1.7 records its placement and every change. No statement of it has a Lean formalization.

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1 What a surreal probability theory should accomplish

A probability can be smaller than every positive real number without being zero. Such a value distinguishes an event that is negligible at ordinary resolution from an event excluded by the model. For example, with $t = \omega^{-1}$, the probabilities

$$(p_1, p_2, p_3) = \frac{(1, t, t^2)}{1 + t + t^2} \quad (1.1)$$

encode three different orders of plausibility. Their ordinary shadow is $(1, 0, 0)$, but conditioning on $\{2, 3\}$ gives $(1/(1+t), t/(1+t))$ on that event. The shadow alone cannot recover this answer. A logit can likewise be positive infinite while its probability remains strictly below one.

The proposal is not to permit probabilities larger than one. Every probability in this article belongs to $[0, 1]$ of an ordered field, hence is a finite surreal. Infinite quantities arise as logits, likelihood ratios, surprisals, utilities, or expectations. Nor is the proposal to replace every ordinary probability calculation: a real probability is already surreal. The added value is the retention of specified, ordered scales of negligible mass and evidence.

1.1 A core theory and several infinite extensions

There are two logically separate tasks. The first is finite probability algebra over an ordered field, augmented by a suitable exponential for information theory. The second is the interpretation of an infinite family of events, observations, or payoffs. The first task has a broad, consistent answer. The second has several useful but inequivalent answers.

Regime	What is required	Appropriate use
Finite algebraic	Positivity, normalization, finite additivity	Exact Bayes, finite decisions, finite graphs and experiments
Strong Hahn	Well-ordered union of supports; finitely many contributions per exponent	Support-controlled infinite algebra and special atomic laws
Coefficientwise	A common well-ordered support of ordinary finite signed measures	Diffuse real laws with infinitesimal layers
Regular finite-additive	A chosen extension, often ultrafilter based	Positive mass for every nonempty event, all-subsets models
Nonstandard/Loeb	Internal spaces and suitable saturation, followed by standard part	Ordinary countably additive limits of hyperfinite experiments

These rows are not five names for the same construction. In particular, “countably additive surreal probability” is incomplete terminology unless the addition and convergence conventions are specified.

1.2 Relation to the supplied repository

The repository was consulted at commit

d22a5b35d5b3040e870c3cfd6d1c8f7259e094b0

Its *Hahn-Valued Measures and Probability* report already distinguishes strong and coefficientwise measures, develops strong atomcity and extension criteria, and studies infinite products and hidden negative mass [13]. Its guide explicitly restricts finite-support standard parts and the support-order threshold $\omega + 1$ to the strong class. We do not generalize those conclusions to other regimes. The foundations report separates formal Hahn sums from fine or valuation

convergence [14]; the Markov report concerns finite generators and scale resolvents rather than general path measures [15].

This article complements those topics by emphasizing conditional shadows, Bayesian and logit geometry, finite information theory, and the interfaces between infinite probability constructions. The two nonextension arguments in Section 12 are elementary special cases consistent with the repository’s broader product discussion; they are not presented as new independent classification theorems. The repository’s maintained guide and the opening of its source were read for the comparison. This is not a line-by-line independent audit of all of its proofs or formalization claims.

Since the pin. The manuscript was placed in the collection by commit 5fe7f8d, seventeen commits after its pin. The statements above remain true of the pin and are kept as provenance. The comparison of Section 1.5 was made at commit 50cb709, against the full sources of the three reports cited above and of the physics report [16]. Between the pin and 50cb709 the source of the measures report did not change. The placement commit added to that report’s directory the program, build script and records of a further manuscript, on coefficientwise measure theory, with files prefixed 16-coefficientwise-measure-theory-; this report makes no statement about that manuscript. The Markov report gained one comparison paragraph about another report, and the foundations and physics reports are unchanged. At 50cb709 a full-text search of every LaTeX source of the collection finds the words `logit`, `softmax`, `relative entropy`, `Bayes`, `martingale` and `Gibbs` only in this report, and `Loeb` elsewhere only in the physics report. Such searches locate words, not results stated under other names.

1.3 Precedents and the meaning of development

The scalar ingredients come from Conway and Gonshor and the exponential-field results of van den Dries and Ehrlich [1, 2, 3, 4]. Set-sized exponential/logarithmic workspaces are also studied explicitly by Bournez and Guilmant [5]. Non-Archimedean probability on infinite spaces, with a nonclassical infinite-addition rule, is developed by Benci, Horsten, and Wenmackers [6]. Halpern and Brickhill–Horsten analyze connections with conditional and lexicographic probability [7, 8]. Hammond already discusses non-Archimedean probability representations and extended logarithmic likelihood ratios [9]. Ordinal ranking theories, including Spohn’s, are an important precedent for scale-sensitive belief revision [10]. Loeb’s theorem provides a different bridge from nonstandard to real measures [11].

Our development consists of precise compatible definitions, detailed consequences, counterexamples, and constructions. It does not depend on asserting historical novelty. Some propositions are useful combinations of standard facts; others are direct calculations specific to this exposition. A proof supplied here establishes the stated implication, not priority over unsearched literature.

1.4 What this report does not claim

Each limitation below is also stated where it applies. They are collected here, from the article, its delivered README, research audit and verification record, so that none of them is lost when results are quoted elsewhere.

- (N1) *Status and priority.* The report is an AI-assisted synthesis with written proofs. It has not been independently refereed, no statement of it has a Lean formalization, and no Lean code accompanies it. Non-Archimedean probability [6], lexicographic and conditional probability [7, 8], extended logarithmic likelihood ratios [9], ranking theories [10] and

Loeb measure [11] are precedents. No first-in-literature result is claimed, no named open conjecture is claimed solved, and no absence claim about the published literature is made. Some propositions are combinations of standard facts; a proof establishes its implication, not priority. A bibliography entry does not certify every theorem of its source. Loeb’s paper was cited from its indexed metadata, since direct access to the full text failed; the Brickhill–Horsten citation is to the preprint; Conway and Gonshor are cited as foundational books; and no ordinal-length estimate of the uncorrected van den Dries–Ehrlich paper is used. Ordinary real disintegration, product laws and strong laws of large numbers are imported from [12], not re-proved.

- (N2) *Repository comparison.* The comparison at the pin read the guide of the measures report and the opening of its source; the descriptions of the foundations and Markov reports rested on the collection’s guides. It was not a line-by-line audit of any report or of the Lean ledger. The strong-class conclusions of the measures report (finitely supported standard part, the $\omega + 1$ threshold) are not generalized to other regimes. The two coin obstructions of Section 12 are special cases of that report’s product theorems (Section 1.5), not new classification theorems.
- (N3) *Scalars and size.* All sample spaces, event algebras, supports and parameter collections are set-sized; no probability distribution over the class **No** is constructed. The closure of Lemma 2.1 supplies no algorithm, no canonical smallest Hahn presentation and no elementary transfer for arbitrary formulas. A Hahn field K_Γ need not be closed under the canonical exponential. The canonical logarithm is not the omega-map, and $\omega^\gamma = \exp(\gamma \log \omega)$ is not assumed for arbitrary γ . Transfer from the real exponential field is used only for universally quantified inequalities in a fixed ordinary number of variables, never for statements quantifying over sample spaces, sequences, measurable functions or integration.
- (N4) *The finite core.* Definition 3.1 carries no countable additivity. Finite Jensen is algebraic convexity, not a statement about a function given only on $\mathbb{R}$ without a specified extension. Zero-mass cells do not determine conditional probabilities. The finite no-sure-loss argument does not license infinitely many tickets, and physical interpretations of infinite stakes are a separate modeling issue.
- (N5) *Shadows and compression.* Standard part forgets small quantities; it does not show that their effect after division by other small quantities is negligible. The lexicographic description after Theorem 4.2 represents conditional shadows only: it does not reconstruct the probability or all expected-utility comparisons. Theorem 4.3 concerns real-valued acts; it does not say that an arbitrary signed-row list is a lexicographic probability system, and it does not compress comparisons of surreal-valued acts.
- (N6) *Bayes and logits.* No unspecified infinite product of likelihoods is licensed; learning from an infinite data stream needs a chosen law on streams and a chosen limit. The exponents of the minimum-plus rule lie in an ordered abelian group, and their addition is not ordinal addition. The endpoints $p = 0, 1$ correspond to formal symbols $\mp\infty$ outside **No**, not to $\mp\omega$. Logits do not add over disjoint events; a score becomes a probability only through a normalization rule; stable normalization gives no finite symbolic representation of an arbitrary surreal score.
- (N7) *Information and optimization.* All sums are finite. The value $+\infty$ of a relative entropy is formal. Entropy can leave a pure Laurent field, and no single Hahn field is assumed closed under the information operations. No entropy rate of a process is supplied, and standard part does not commute with relative entropy. The logistic optimum comes from an explicit identity, not an extreme-value theorem. The Gibbs principle constructs no

thermodynamic limit, infinite-volume Gibbs measure or partition function on an infinite state space. The standard part of an expected utility needs limited payoffs, and surreal arithmetic does not decide which utilities and trades are admissible. The smoothing constants and scale are modeling choices, and the case $N = 0$ is an ordinary prior. Proposition 8.2 gives no existence theorem for constrained or regularized regression.

- (N8) *Finite processes.* Consistency of finite-horizon path weights is not a law on infinite paths; a stochastic matrix, a semigroup and a path measure are different objects. Optional stopping is proved for bounded stopping times only. “Take $n = 1/t$ ” has no meaning until a nonstandard or surreal-indexed number of trials is defined. No unconditional law of large numbers in the fine topology is hidden in the finite variance bounds.
- (N9) *Infinite addition.* Proposition 10.1 concerns the full fine topology and set-sized index sets, not the intrinsic topology of a subfield. Failures of the strong rule are failures of a support rule, not contradictions in the field or in finite additivity. The strong classification of the measures report is quoted, not re-proved. Positivity on cylinders or finite tests does not imply positivity on all measurable sets.
- (N10) *Hierarchies and integration.* Setting $t = 1$ in a hierarchy is not an ordinary mixture. Dominated convergence holds coefficientwise, not in the fine or order topology. The Fubini identity is support-controlled and does not cover all surreal-valued functions. Theorem 11.3 asserts coefficientwise measurability only and does not contradict Lebesgue integration; the countable-support positivity statement after it is a sufficient condition. The posterior kernels of Section 11.4 are not a general surreal Radon–Nikodym or disintegration theorem; their version choices remain real choices, and no unique conditional law is assigned to observations of exactly zero mass. Example 11.2 is not an all-subsets regular probability.
- (N11) *Coin obstructions.* Theorems 12.1 and 12.2 do not make any finite coin experiment inconsistent, do not exclude a regular finitely additive path law (Corollary 13.3 constructs one), and do not say that every non-Archimedean Kolmogorov-type theorem fails. A logit parametrization supplies no infinite-addition rule.
- (N12) *Extensions.* The embeddings of Lemma 13.1 need not preserve an exponential, internal-set structure or a Hahn summation, and do not show that two exponential structures agree. The ultrafilter construction is nonconstructive, its snapshot addition is not a strong sum, the resulting laws are not unique even with a fixed infinitesimal unit, and symmetry is not implied by equal singleton masses. Theorem 13.2 provides no countable additivity, symmetry, canonical embedding or bound on the new scales, and does not compute the probability of a tail event, which can depend on the model chosen. Its compactness is first-order logical compactness, not topological compactness of a simplex.
- (N13) *Latent regime and bridges.* Proposition 14.1 is consistent with the real martingale convergence theorem; a surreal analogue must specify a different conclusion or further hypotheses. Writing an infinite logit does not define a surreal-indexed sum of log-likelihood increments. The Loeb bridge needs internal sets and saturation; a field embedding does not supply them, and a bare surreal $H > \mathbb{N}$ is not a substitute. An internal power is not identified with a canonical surreal exponential expression under a field-only embedding.
- (N14) *Not asserted anywhere* (Section 17.2). No countably additive probability on all events for the full fine topology; no universal Carathéodory, Radon–Nikodym, Fubini, martingale convergence, strong law of large numbers, central limit or stochastic calculus theorem after replacing $\mathbb{R}$ by $\mathbf{No}$; no unique answer to exact-zero conditioning; no canonical

ultrafilter; no reconciliation of regularity with every symmetry demand; no effective computability of arbitrary surreal expressions; no identification of the omega-map with exponential powers; no transfer of internal hyperfinite operations through a field-only embedding. The Poisson and latent examples import ordinary component laws or a stated nonstandard model.

- (N15) *Computation and evidence.* The engine of Section 16 is a specification; a leading-scale answer is not an exact probability. The program `code/verify.py` checks finite examples in $\mathbb{Q}(t)$ with exact rational arithmetic and no floating-point surrogate for t . It implements no canonical exponential or logarithm, no infinite summation, measure extension, saturation, compactness, choice or uncountable set, and it proves no general theorem. Compilation and visual inspection are typesetting checks, not proof checks.
- (N16) *Open.* The four directions of Section 17 are not settled: positive integration domains for coefficientwise probabilities beyond bounded real observables and countable supports; convergence of conditional beliefs that keeps chosen infinitesimal regimes; extension with additional structure (invariance, a preferred support group, an exponential-compatible embedding); and a verified computational library.

1.5 Relation to other reports of the collection

No proof in this report depends on another report, and no other report depends on this one. The comparisons below are with the sources as committed at 50cb709; theorem numbers of other reports are those of their builds at that commit, and each is followed by its label.

Hahn-valued measures and probability. That report [13] is the collection’s measure theory for Hahn-valued probabilities. This report continues it on the finite side and at its interfaces with other infinite constructions; it answers none of the questions of Section 33 there.

- (a) *The same definitions.* Definition 10.3 is Definition 2.7 there (meas:def:coefficientwise), with the probability condition of Definition 2.9 there (meas:def:positive). The strong rule of Section 10 is Definition 2.8 there (meas:def:strong), built on the strong summability of Definition 2.2 there (meas:def:summability). Finite additivity (Definition 3.1 here) is the third notion, and neither report identifies it with the other two. On a finite event algebra a disjoint sequence has only finitely many nonempty members, so all three notions coincide there; this is why Sections 4 and 5 may speak of a *finite Hahn probability* without a qualifier.
- (b) *Strong class.* Example 10.2 is an instance of the atomic classification, Theorem 4.2 there (meas:thm:atomic). The “classification” quoted in Section 10 is that theorem with Corollaries 4.5 and 4.6 there (meas:cor:two-axioms, meas:cor:shadow): on countably separated spaces strong measures are the coefficientwise ones with finitely atomic coefficients, and a strong probability has a finitely supported standard part. The Lebesgue remark of Section 10 is Trap 2 of Section 2.6 there (meas:sub:traps).
- (c) *The coin obstructions.* Theorem 12.1 (bias t) and Theorem 12.2 (bias $1/2 + t$) are two cases of Corollary 23.1 there (meas:cor:iid): the i.i.d. Bernoulli law with bias p has a coefficientwise *signed* extension if and only if p is real. That corollary rests on the total-variation extension test, Theorem 19.1 there (meas:thm:extension). Its proof uses the same two mechanisms: the first infinitesimal coefficient of bias t^α has level- n variation $2n$, where Theorem 12.1 uses the first-success events, and for an interior bias the variation grows like $\sqrt{n}$ by Hölder’s inequality and fourth-power expectations, Lemma 20.3 there (meas:lem:L1lower), as in the proof of Theorem 12.2. Neither coin law has a strong extension either (Corollary 10.3 and Example 10.4 there, meas:cor:bernoulli, meas:ex:bernoulli). Corollary 13.3 adds what that

report leaves aside: its non-claims 10.1 and 15.1 (Section 32.2 there, meas:sub:nonclaims) say that its negative results do not exclude finitely additive or ultrafilter probability, and Corollary 13.3 gives both coin models regular finitely additive laws on all subsets.

- (d) *Conditioning.* Proposition 24.2 there (meas:prop:conditioning) shows that conditioning a coefficientwise probability on an event of positive mass stays in the class, and Example 24.3 there (meas:ex:conditioning) computes a conditional shadow with infinitely many atoms. Theorem 11.1 computes every conditional shadow of a normalized hierarchy, and Section 11.4 conditions on continuous observations; neither result is in that report.
- (e) *Observables.* The expectation (11.4) is the restriction to normalized hierarchies of the integral of bounded ordinary observables in Section 24 there (meas:eq:coef-integral, defined for every coefficientwise measure), with the same coefficientwise dominated convergence. That report says that integrating arbitrary Hahn-valued functions would need further joint support and integrability conditions and asserts nothing about it. Theorem 11.3 shows that a common well-ordered support with bounded integrable coefficients is not enough at uncountable support rank, and the paragraph after it shows that a countable support is enough for positivity. Its mechanism, an uncountable union of null sets, is reminiscent of the null-ideal criterion, Theorem 19.2 there (meas:thm:nullideal), in which positivity of a coefficientwise measure is decided on the null ideal common to all earlier coefficients, and this ideal is that of one control measure when the predecessor set is countable. The two results concern different objects (integrals of Hahn-valued functions against an ordinary measure here, event masses of a Hahn-valued measure there), and neither is derived from the other.
- (f) *Open questions.* Question 33.9 there (meas:q:markov), dependence governed by a transition kernel, stays open: Section 9 has finite horizons only, and the latent model of Section 14 is an exchangeable mixture of two product laws.

Foundations. The strong summability used here is (10.1) of the foundations report [14] (found:eq:summability), and the positive-support Neumann lemma is its Section 10.2 (found:sub:positivesupport). Proposition 10.1 is clause (2) of Theorem 12.1 there (found:thm:discrete), with the same proof; that theorem also makes set-sized subsets discrete and small-index Cauchy nets eventually constant. The distinctions drawn in Section 10 between the full fine topology, the intrinsic topology of a subfield and coefficientwise real convergence are those of its Section 12 (found:sec:topology) and of the topology table in docs/NOTATION.md.

Markov generators at every scale. The Markov report [15] studies the normalized resolvent $R_L(s) = s(sI + L)^{-1}$ of a finite rate matrix over a Hahn field with divisible value group. Its non-claim (N1) (Section 1.4 there, markov:sec:nonclaims) excludes path measures, sample paths and surreal-valued probabilities on path space. Section 9 builds finite-horizon path weights from any finite stochastic matrix over K . The following link is an observation of the placement, not of either source. By Proposition 3.1 there (markov:prop:forest), for positive s the matrix $R_L(s)$ is entrywise positive and row stochastic with limited entries. So it may serve as T in Section 9, and there the path shadow is the ordinary finite-horizon chain whose transition matrix is the residue matrix $K_\alpha(c)$ of $R_L(ct^\alpha)$, which that report calls the shadow at scale α . This gives laws of finite paths only; neither report constructs a law on infinite paths. The relative-stability Theorem 8.1 there (markov:thm:stability) concerns resolvent entries under relative rate errors, and Theorem 5.3 concerns conditional probabilities under absolute or valuation-controlled errors; they are statements about different maps.

Surreal scalars and spacetime. The probability subsection of the physics report [16] (Section 15.4 there, `phys:sub:probability`) observes that equal positive infinitesimal weights on infinitely many outcomes are not strongly summable (Example 4.3 there, `phys:ex:summable`); that a finite independent-trial model with a fixed ordinary number of trials keeps an infinitesimal event infinitesimally likely; and that non-Archimedean theories [6] and Loeb’s construction [11] need additional structure and an account of how records relate to field-valued weights. Section 10 repeats the first point, (9.3) is the second, the fine-ultrafilter law of Section 13 gives equal positive singleton masses by snapshot addition, not by a strong sum, and Section 15 states what Loeb’s construction needs. Like that report, this one gives no physical interpretation of field-valued probabilities.

1.6 Words and symbols used differently elsewhere in the collection

Several words and letters of this report mean other things elsewhere. Each has one meaning here.

- *Additivity.* *Finite additivity* is Definition 3.1. *Strong* additivity is the rule of Section 10, and *coefficientwise* additivity is Definition 10.3; both are the measures report’s classes. The snapshot addition of an ultrapower (Section 13) and Loeb’s countable additivity of a real standard part (Section 15) are further, different rules. *Strong* always refers to strong Hahn summability; the classical strong law of large numbers is always written in full.
- *Regular* is Brickhill and Horsten’s regularity: every nonempty event of the algebra has positive probability. It is not Radon regularity, and the measures report gives the word no meaning of its own.
- *Limited* means bounded in magnitude by an ordinary integer, which `docs/NOTATION.md` calls *finite*. Here *finite* counts things (finitely many states, terms, trials or tickets); an “ordinary finite n ” is a standard natural number, and the “finite surreal” of Section 1 is a limited one.
- *Shadow.* $P_0 = \text{st } P$, the conditional shadow $C(A \mid B)$ and the path shadow are standard parts, `st` being the collection’s scalar standard part. The Markov report’s shadow is the same operation applied to a resolvent matrix. No standard-part topology is used.
- *Fine.* The fine topology is the collection’s full surreal fine topology. A *fine* ultrafilter (Sections 13.2 and 13.3) is one containing every cone $\{s : F \subseteq s\}$; that sense, from the non-Archimedean probability literature, has nothing to do with the topology.
- *Hierarchy* means a normalized hierarchy of ordinary probability laws (Theorem 11.1), not the Markov report’s hierarchy of stochastic projections.
- *Moment.* The fourth moment $\mathbb{E}S_n^4$ in the proof of Theorem 12.2 is the expectation of a power of an ordinary random variable, which the measures report calls a fourth-power expectation; no moment problem appears here.
- *Positive*, for a Hahn series, means a positive leading coefficient. A positive coefficientwise probability is nonnegative on every event, as in the measures report; positivity of every event is regularity.
- *Local letters.* H is an entropy $H(p)$ (Section 7), a hypothesis with data D in (6.2), a hyperinteger sample size (Sections 13.2, 13.3 and 15.3) and the latent label of Section 14. E is an exponential–logarithmic workspace, and $E_i \in E$ are energies in Theorem 8.1; the blackboard $\mathbb{E}$ is expectation. λ_γ is $-\log t^\gamma$, while λ alone is a precision in Theorem 5.3, Lebesgue probability in Example 11.2 and a Poisson parameter in Section 15. D is a relative entropy $D(p\|q)$, a leading constant in Theorem 5.1 and a density $D(y)$ in (11.8). T is a

stochastic matrix in Theorem 7.2 and Section 9.1 and a stopping time in Section 9.2. K is the scalar field, K_j are conditional kernels and K_n are counts in Section 14. $C(A | B)$ is a conditional shadow and C a real constant in Section 5. In examples $t = \omega^{-1}$.

- The valuation is $v(x) = \min \text{supp } x$, lc is the leading coefficient, and the surreal embedding is $t^\gamma \mapsto \omega^{-\gamma}$, as elsewhere in the collection. $\exp$ and $\log$ are the canonical ordered surreal exponential and its inverse, never a global surcomplex exponential.

1.7 Provenance of this report

This report is built from one manuscript, *Surreal Probability and Log-Odds: A multiscale theory of belief, information, and infinite sampling*, dated 22 September 2026. It arrived as the archive `surreal_probability.zip`, number 05 of nine archives committed on arrival at `2765c8f`. The archive held the LaTeX source `surreal_probability.tex`, its PDF, a README, the research audit `RESEARCH_AUDIT.md`, the program `verify.py` with its recorded output `verification.json`, a build record `BUILD_VALIDATION.json` and a build script. It was placed at commit `5fe7f8d`. Its repository comparison is pinned to `d22a5b35d5b3040e870c3cfd6d1c8f7259e094b0` (Section 1.2).

There is no second source, so there was no merge and no choice between sources. Placement retained every theorem, proof, example and disclaimer of the manuscript. The manuscript continues the measures report but answers no question it names, so it was placed as a report of its own rather than as a part of that one.

Placement changed the following and nothing else.

- The source was renamed `article.tex`. The program and build script were moved to `code/`, the two JSON records to `data/`; they and `RESEARCH_AUDIT.md` are byte-identical to the delivered files. The delivered PDF and README are not shipped.
- All 100 labels of the manuscript received the prefix `fsp:`, and every reference was updated. Fourteen labels were added, for the new subsections and for the definition, subsections and appendix that the new text cites.
- Sections 1.4–1.7, the paragraph “Since the pin” in Section 1.2 and pointer sentences in Sections 10 and 12 were added. They contain no numbered statement or equation, so every theorem, equation and section number of the manuscript is unchanged. The title-page status paragraph now states the draft status and the placement.
- To keep one meaning of *strong*, the title of Example 10.2 (“A regular strong law”) now reads “A regular strong Hahn probability”, and the four mentions of classical strong laws (Sections 14 and 17.2, Appendix A) now read “strong law(s) of large numbers”.
- Section 16.3 and Appendix B now describe the files as placed and record a rerun of the checks; the bibliography entries of the repository reports carry a placement note, and an entry for the physics report was added.

Placement left the mathematics unchanged. The subsequent proof review of Sections 2–3 expands the elementary arguments and supplies the full-event-algebra hypothesis for the point-weight representation. On a smaller finite algebra it is the measurable atoms, not necessarily the points, that carry the uniquely determined weights. It also makes the limited-input and conditional-version conventions explicit. The review of Sections 4–5 subsequently adds the joint-normalizer hypothesis for commuting likelihood updates, corrects the neutral-likelihood wording, and separates sufficient conditioning precision from necessity. The live review record is `docs/REVIEW.md`. The Sections 6–7 review makes the finite nonempty domains and binary-prior hypotheses explicit, expands softmax and information inequalities, and separates an attained

interior logarithmic-score minimum from the boundary infimum. The delivered audit, programs and data remain unchanged. The Sections 8–9 pass adds the posterior-support qualification after smoothing, expands the Gibbs and finite-process proofs, and makes adaptation and stopping-time measurability explicit. The Sections 10–11 pass expands common-support and integration arguments, distinguishes uniform bounds from one-component essential bounds, and supplies a posterior with nonintegrable coefficients before density cancellation. The Sections 12–13 pass expands the coefficient-variation obstructions, the relative field embedding and finite compactness models, and replaces the logit change-of-variable shortcut by a calculation in the original Hahn coordinates.

2 Scalars, size, and the two meanings of exponentiation

2.1 Set-sized workspaces inside the surreals

The surreal numbers $\mathbf{No}$ form a proper class, not an ordinary set-valued sample range. We use set-sized sample spaces, event algebras, supports, and collections of parameters. The algebraic field K always denotes a set-sized ordered subfield of $\mathbf{No}$ containing the displayed copy of $\mathbb{R}$. When logarithms are used we work in a set-sized subfield $E \subseteq \mathbf{No}$ closed under the canonical surreal exp and under log on positive arguments. In a class theory these are sets inside the ambient class; in a universe formulation all sizes are interpreted relative to the chosen universes. No probability distribution over the entire proper class $\mathbf{No}$ is constructed.

Lemma 2.1 (Local exponential workspace). *Every set $S \subseteq \mathbf{No}$ is contained in a set-sized subfield E containing $\mathbb{R}$ and closed under canonical exponential and positive logarithm.*

Proof. Let E_0 be the field generated by $\mathbb{R} \cup S$. Given E_n , let E_{n+1} be the field generated by $E_n \cup \exp(E_n) \cup \log((E_n)_{>0})$. Replacement and finite field expressions make each stage a set. The union $E = \bigcup_{n < \omega} E_n$ has the required properties because each input and each finite list of inputs already lies at one stage. More explicitly, the stages form an increasing chain, so two elements of E lie together in some E_n . Their sum, product and any defined quotient lie in that field. For $x \in E_n$, its exponential and, when $x > 0$, its logarithm lie in E_{n+1} . This proves each claimed closure separately. $\square$

This lemma supplies closure, not an algorithm, a canonical smallest Hahn presentation, or elementary transfer for arbitrary quantified formulas. The particular scalar inequalities used below hold in $\mathbf{No}$ and therefore for their inputs in E .

2.2 Standard part

An element x is *limited* if $|x| < n$ for some ordinary positive integer n , and *infinitesimal* if $|x| < 1/n$ for every such n . Write $x \simeq y$ when $x - y$ is infinitesimal. Every limited surreal has a unique real standard part, denoted $\text{st } x$, with $x \simeq \text{st } x$. Indeed, if $|x| < n$, the set $\{a \in \mathbb{R} : a < x\}$ is nonempty and bounded above in $\mathbb{R}$. Let r be its real supremum. For each real $\delta > 0$, the supremum property gives $r - \delta < x$, while $r + \delta/2$ exceeds the supremum and hence cannot be strictly below x . Hence $r - \delta < x < r + \delta$. This proves $x \simeq r$. Two such real numbers differ by a real infinitesimal and therefore agree. The Dedekind completeness used here belongs to $\mathbb{R}$, not to $\mathbf{No}$. Here *limited* means what the collection's notation guide calls a *finite element*; it is unrelated to finite birthday or finite support.

Proposition 2.2 (Residue algebra). *On limited elements, standard part is an order-preserving ring homomorphism. If x, y are limited and $\text{st } y \neq 0$, then x/y is limited and $\text{st}(x/y) = \text{st } x / \text{st } y$. A limited x is infinitesimal exactly when $\text{st } x = 0$.*

Proof. Write $x = r + \eta$ and $y = s + \theta$ with $r, s \in \mathbb{R}$ and infinitesimal η, θ . Sums and products of infinitesimals with limited factors are infinitesimal. If $s \neq 0$, then y is bounded away from zero by a positive real, and $y^{-1} - s^{-1} = -\theta/(sy)$ is infinitesimal. Thus y^{-1} is limited with standard part s^{-1} , proving the quotient assertion. If $x \leq y$ but $r > s$, the positive real gap $r - s$ cannot be canceled by the infinitesimal $\theta - \eta$, a contradiction. Finally $x = r + \eta$ is infinitesimal if and only if $r = 0$. The map fixes real constants, so it is onto $\mathbb{R}$; its kernel is precisely the infinitesimal ideal of the limited-element ring. It is not strictly order preserving: $0 < t = \omega^{-1}$ but $\text{st } t = \text{st } 0 = 0$. $\square$

A field need not contain all limited surreals to have this residue map; containing $\mathbb{R}$ is sufficient for the standard parts of its own elements. Taking standard part forgets small quantities; it does not prove that their statistical or decision-theoretic effects are negligible after division by other small quantities.

2.3 Hahn coordinates and leading scales

For explicit expansions let Γ be a set-sized ordered additive subgroup of $\mathbf{No}$, and put

$$K_\Gamma = \mathbb{R}((t^\Gamma)), \quad t^\gamma \mapsto \omega^{-\gamma}. \quad (2.1)$$

The exponent support is well ordered in increasing order. These normal forms identify K_Γ with an ordered subfield of $\mathbf{No}$ [3, Section 2, p. 176]; the field need not be closed under the canonical exponential. An abstract ordered group can instead be used with a specified embedding into the additive surreals. For $x \neq 0$ set

$$v(x) = \min \text{supp}(x), \quad \text{lc}(x) = [t^{v(x)}]x, \quad v(0) = +\infty. \quad (2.2)$$

Here $+\infty$ is a bookkeeping symbol for the valuation, not a surreal. The order of a nonzero series is the sign of its leading coefficient. For positive x, y ,

$$v(xy) = v(x) + v(y), \quad v(x + y) = \min\{v(x), v(y)\}. \quad (2.3)$$

For products the least exponent is $v(x) + v(y)$, with coefficient $\text{lc}(x)\text{lc}(y) \neq 0$. For sums with unequal valuations, the smaller exponent survives; with equal valuations its coefficient is $\text{lc}(x) + \text{lc}(y) > 0$. This proves both identities without a cancellation assumption on any higher coefficient. If $v(p) < 0$ and $p > 0$, its leading monomial dominates every positive real constant, contradicting $p \leq 1$. Thus probabilities have $v(p) \geq 0$. More generally a Hahn element is limited exactly when its valuation is nonnegative, and its standard part is then $[t^0]x$; zero has the assigned valuation $+\infty$.

A family $(x_i)_{i \in I}$ is *strongly Hahn summable* if its combined support is well ordered and, for each exponent, only finitely many members have a nonzero coefficient there. Its sum, written $\sum_i^H x_i$, is formed coefficientwise using those finite sums. The standard positive-support Neumann lemma justifies geometric, exponential, and logarithmic formal series in infinitesimal Hahn arguments; it is a support theorem, not a statement about ordinary sequence limits [2, 14].

2.4 Canonical logarithm is not the omega-map

The canonical surreal exponential is an order-preserving group isomorphism from $(\mathbf{No}, +)$ to $(\mathbf{No}_{>0}, \cdot)$ extending real exponential [2, 3]. Its inverse is our only meaning of log. The omega-map $\gamma \mapsto \omega^\gamma$ that defines Conway monomials is a different operation. In particular, one must not substitute

$$\omega^\gamma = \exp(\gamma \log \omega) \quad \text{for arbitrary surreal } \gamma$$

without an additional theorem on the exponents in question. A safe log-scale notation is

$$\lambda_\gamma = -\log(t^\gamma). \quad (2.4)$$

It satisfies $\lambda_{\gamma+\delta} = \lambda_\gamma + \lambda_\delta$ and is strictly increasing. Thus, for $c > 0$ real and infinitesimal u ,

$$\log(ct^\gamma(1+u)) = \log c - \lambda_\gamma + \log(1+u). \quad (2.5)$$

Integer powers of a fixed t are ordinary field powers and cause no such ambiguity.

We use the scalar inequalities

$$\exp x \geq 1 + x, \quad \log y \leq y - 1 \quad (y > 0), \quad (2.6)$$

with equality only at $x = 0$ and $y = 1$, respectively. These follow from the elementary-extension theorem [3, Corollary 2.2, p. 177]. The erratum [4] corrects ordinal support bounds in Section 4 and leaves this transfer statement unchanged. No such ordinal bound is used here. For each fixed ordinary integer n , a universally quantified real-exponential inequality involving n coordinates can be used this way: first it holds in **No**, then its quantifier-free instance at inputs in E has the same meaning there, since E inherits the order and field operations and is closed under the displayed exponential and logarithm. The workspace lemma alone does not make E an elementary substructure. This does *not* transfer a theorem quantifying over arbitrary sample spaces, countable sequences, measurable functions, or integration operations. The distinction will matter repeatedly.

3 The finite probability core

3.1 Events, weights, and random variables

Definition 3.1. A K -probability on an algebra $\mathcal{A}$ of subsets of a nonempty set Ω is a function $P : \mathcal{A} \rightarrow K$ such that $P(\Omega) = 1$, $P(A) \geq 0$, and

$$P(A \cup B) = P(A) + P(B) \quad \text{if } A \cap B = \emptyset. \quad (3.1)$$

It is *regular* if $P(A) > 0$ for every nonempty $A \in \mathcal{A}$. No countable-additivity assertion is implicit in this definition.

Taking the two disjoint events to be empty gives $P(\emptyset) = 0$. The decompositions $\Omega = A \sqcup A^c$ and $B = A \sqcup (B \setminus A)$ when $A \subseteq B$ give complements and monotonicity, hence $0 \leq P(A) \leq 1$.

On $\Omega = \{1, \dots, n\}$ with the full event algebra $\mathcal{A} = 2^\Omega$, a law is equivalent to its point-weight vector $p_i = P(\{i\}) \geq 0$, with $\sum_i p_i = 1$; finite additivity gives $P(A) = \sum_{i \in A} p_i$, and this formula conversely defines a law. Regularity means exactly that all $p_i > 0$. For a smaller finite event algebra, use its nonempty minimal events (its *atoms*) instead: these partition Ω , every event is a union of them, and their masses uniquely determine the law. Individual point masses need not be defined. For example, on $\{1, 2\}$ with algebra $\{\emptyset, \Omega\}$, the point-weight vectors $(1, 0)$ and $(0, 1)$ restrict to the same law. Throughout the finite vector models below, all subsets are measurable; this is an additional convention, not a consequence of finiteness alone.

A finite-range K -valued random variable is a function constant on the members of a finite measurable partition. Its expectation is

$$\mathbb{E}_P X = \sum_j x_j P(A_j), \quad X = \sum_j x_j \mathbf{1}_{A_j}. \quad (3.2)$$

For two representations, pass to the common partition of nonempty intersections of their cells. On each such intersection the two assigned values agree, and finite additivity splits each original

mass into the masses of these intersections. Thus the displayed sum is independent of the representation. A common partition also proves K -linearity and positivity. Applying positivity to $X - \min X$ and $\max X - X$, and using $\mathbb{E}1 = 1$, gives $\min X \leq \mathbb{E}X \leq \max X$. No convergence notion is needed, even if the finitely many values are infinite surreals. If $P(B) = 0$, then $\mathbb{E}[X\mathbf{1}_B] = 0$: every cell inside B has mass zero, whatever its assigned value.

For two finite laws, the product weights $p_i q_j$ define independent coordinates. A finite map $f : \Omega \rightarrow S$ pushes the law forward by $P_f(s) = \sum_{f(i)=s} p_i$. Every ordinary finite marginalization, finite inclusion–exclusion identity, and finite factorization argument therefore works verbatim as field algebra. A finite acyclic graphical model with stochastic local tables defines a normalized joint law by multiplying those tables; normalization follows by summing terminal vertices in a reverse topological order. This is an actual construction, not an appeal to a missing infinite extension theorem.

3.2 Finite inequalities

Theorem 3.2 (Variance, Cauchy–Schwarz, and tail bounds). *For finite-range measurable variables X, Y on the same K -probability space, define $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$. Then*

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] \geq 0, \quad (3.3)$$

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2], \quad (3.4)$$

$$P(X \geq a) \leq \mathbb{E}[X]/a \quad (X \geq 0, a > 0), \quad (3.5)$$

$$P(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2 \quad (a > 0). \quad (3.6)$$

The thresholds and variable values may be surreal.

Proof. Work on a common finite measurable partition for X and Y . Positivity of squares gives the variance assertion, and expansion gives $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$. Put $A = \mathbb{E}[X^2]$, $B = \mathbb{E}[XY]$, and $C = \mathbb{E}[Y^2]$. If $C = 0$, each nonnegative summand $p_j y_j^2$ is zero, so every positive-mass cell has $y_j = 0$ and $B = 0$. If $C > 0$, expansion at $c = B/C$ gives

$$0 \leq \mathbb{E}[(X - cY)^2] = A - 2cB + c^2C = A - B^2/C.$$

Multiplication by C proves (3.4). For Markov, $a\mathbf{1}_{\{X \geq a\}} \leq X$ pointwise; apply positivity of expectation and divide by $a > 0$. For Chebyshev, apply Markov to $(X - \mathbb{E}X)^2$ at the positive threshold a^2 . The events agree because $|z| \geq a$ if and only if $z^2 \geq a^2$ in an ordered field. No division by a possibly zero second moment or probability is used. $\square$

If a function f is convex on an interval of K in the ordered-field sense, finite Jensen follows by induction on the number of positive weights. With at least two such weights, separate one of weight $p \in (0, 1)$, normalize the remaining weights by $1 - p$, apply the inductive assertion, and then the defining two-point convexity inequality. One positive weight is the equality case; zero weights may be discarded. One cannot apply this assertion to an arbitrary function originally defined only on $\mathbb{R}$ without first specifying its extension to K . The result is algebraic convexity, not a general surreal integration theorem.

3.3 Conditioning and finite conditional expectation

Whenever $P(B) > 0$, define

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}. \quad (3.7)$$

An infinitesimal denominator is nonzero and poses no algebraic problem. The map $A \mapsto P(A \mid B)$ is again a finitely additive law. For $P(A), P(B) > 0$, cancellation gives $P(A \mid B)P(B) = P(B \mid A)P(A)$; division by $P(B)$ gives Bayes' rule. Likewise $P(A \cap B \mid C) = P(A \mid B \cap C)P(B \mid C)$ whenever $P(B \cap C) > 0$ (which also ensures $P(C) > 0$). Each iterated chain rule requires the displayed conditioning masses to be positive; an exact zero does not define a quotient. For a finite partition (B_j) with positive masses,

$$P(A) = \sum_j P(A \mid B_j)P(B_j). \quad (3.8)$$

Zero-mass cells, when present, contribute zero and may be omitted; assigning arbitrary conditional values there does not determine a unique conditional probability system on those cells.

For a finite subalgebra $\mathcal{G}$ with positive-mass atoms, set $\mathbb{E}[X \mid \mathcal{G}]$ on an atom B equal to $\mathbb{E}[X\mathbf{1}_B]/P(B)$. Direct finite summation gives

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}X, \quad \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}] \quad (\mathcal{H} \subseteq \mathcal{G}). \quad (3.9)$$

For the first identity, summing over the atoms cancels their positive masses. For the second, let C be an atom of $\mathcal{H}$ and sum over the atoms B of $\mathcal{G}$ contained in C :

$$\sum_{B \subseteq C} \frac{\mathbb{E}[X\mathbf{1}_B]}{P(B)} P(B) = \mathbb{E}[X\mathbf{1}_C].$$

Dividing by $P(C)$ gives the equality on C . A $\mathcal{G}$ -measurable variable Z is constant on each atom, so $\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}]$ there.

Write $M = \mathbb{E}[X \mid \mathcal{G}]$ and define $\text{Var}(X \mid \mathcal{G}) = \mathbb{E}[(X - M)^2 \mid \mathcal{G}]$. On each atom the defining average gives $\mathbb{E}[(X - M)\mathbf{1}_B] = 0$, hence $\mathbb{E}[(X - M)Z] = 0$ for every such Z . Taking $Z = M - \mathbb{E}X$ in the square expansion $X - \mathbb{E}X = (X - M) + (M - \mathbb{E}X)$ proves the finite total-variance identity

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid \mathcal{G})] + \text{Var}(\mathbb{E}[X \mid \mathcal{G}]). \quad (3.10)$$

If some atoms have zero mass, assign arbitrary constant values on them. The conditional identities then hold outside a measurable null event; scalar expectation and variance identities remain exact and independent of those choices, since null cells contribute zero. Thus uniqueness is pointwise when every atom has positive mass, and otherwise only up to these versions. None of this requires bounded real payoffs: the values need only form a finite list in K .

3.4 A finite betting interpretation

A ticket on A paying $\mathbf{1}_A$ and priced at $P(A)$ has centered gain $\mathbf{1}_A - P(A)$. Any finite K -linear combination of such gains has expectation zero. It cannot be strictly negative in every state: on the finite partition generated by the tickets, a strictly negative value on every nonempty cell gives a strictly negative expectation, since at least one cell has positive mass. This is a finite no-sure-loss statement with K -valued stakes and payoffs. It does not license infinitely many tickets or an infinite series of stakes.

Conversely, write a quoted price as $q(A)$ and its centered ticket as $T_A = \mathbf{1}_A - q(A)$. If $q(\Omega) \neq 1$, then T_Ω is a nonzero constant. If disjoint events violate finite additivity, then

$$T_{A \cup B} - T_A - T_B = q(A) + q(B) - q(A \cup B)$$

is a nonzero constant. Choosing its sign gives a strict sure loss. If $q(A) > 1$, buying T_A loses in every state; if $q(A) < 0$, selling it does. Thus absence of a strict sure loss forces precisely normalization, nonnegativity and finite additivity. It does not force regularity: on a two-point full event space, the weights $(1, 0)$ are coherent although one nonempty event has price zero. Physical interpretations of infinite stakes are a separate modeling issue; finite coherence alone does not require such stakes to be practically available.

4 Conditional shadows and finite scale hierarchies

4.1 The unconditional shadow is not enough

Since every event mass lies in $[0, 1]$, standard part is defined on it. Its positivity, additivity and value at 1 show that $P_0(A) = \text{st } P(A)$ is a real finitely additive probability. For $P_0(B) > 0$, Proposition 2.2 gives

$$\text{st } P(A \mid B) = P_0(A \mid B). \quad (4.1)$$

For $P_0(B) = 0 < P(B)$ the right-hand side is undefined, and no replacement can be inferred from P_0 alone.

Example 4.1 (Identical shadows, different rare-event answers). On three states let

$$p = (1 - 2t, t, t), \quad q = (1 - 4t, t, 3t), \quad B = \{2, 3\}, \quad t = \omega^{-1}. \quad (4.2)$$

Both shadows are $(1, 0, 0)$, whereas $P(\{2\} \mid B) = 1/2$ and $Q(\{2\} \mid B) = 1/4$. The coordinate differences are $(2t, 0, -2t)$, so their total variation distance is $\frac{1}{2}(2t + 0 + 2t) = 2t$.

For a regular probability on an event algebra $\mathcal{A}$ there is a richer real object, defined for $A, B \in \mathcal{A}$ with $B \neq \emptyset$:

$$C(A \mid B) = \text{st } \frac{P(A \cap B)}{P(B)}, \quad B \neq \emptyset. \quad (4.3)$$

For each fixed B it is a real finitely additive probability concentrated on B . Whenever $D \neq \emptyset$ and $B \cap D \neq \emptyset$,

$$C(A \cap B \mid D) = C(A \mid B \cap D)C(B \mid D). \quad (4.4)$$

Regularity makes $P(D)$ and $P(B \cap D)$ positive, so the ordinary cancellation identity for these ratios applies before standard part. Each factor lies in $[0, 1]$, and the ring law for standard part gives (4.4), even when $C(B \mid D) = 0$. The conditioning domain is precisely $\mathcal{A} \setminus \{\emptyset\}$. This has the normalization, finite-additivity and product properties of conditional probability spaces in [7, Definition 2.1]; compare also [8]. No convention for the empty conditioning event is needed. The existence of the ratios comes from regularity of P , whereas the real conditional law itself can vanish on nonempty events.

4.2 Leading atoms determine every conditional real shadow

Theorem 4.2 (Conditional leading-scale skeleton). *Let a finite regular Hahn probability have atoms indexed by $\{1, \dots, n\}$ and weights*

$$p_i = c_i t^{a_i} (1 + \eta_i), \quad c_i > 0, \quad v(\eta_i) > 0, \quad (4.5)$$

where $c_i \in \mathbb{R}_{>0}$, $a_i \in \Gamma$, and a zero error is allowed by $v(0) = +\infty$. Identify events with their sets of atom indices. For a nonempty event B , put $a(B) = \min_{i \in B} a_i$. Then

$$\text{st } P(A \mid B) = \frac{\sum_{i \in A \cap B, a_i = a(B)} c_i}{\sum_{i \in B, a_i = a(B)} c_i}. \quad (4.6)$$

Consequently all conditional shadows are determined by the ordered partition of atoms into equal- a_i groups and their positive leading coefficients, up to a common positive rescaling within each group.

Proof. Divide numerator and denominator of (3.7) by $t^{a(B)}$. Each resulting summand has the form $c_i t^{a_i - a(B)}(1 + \eta_i)$. It is limited: its residue is c_i when $a_i = a(B)$, and zero when $a_i > a(B)$. Hence the normalized denominator has positive real residue $\sum_{i \in B, a_i = a(B)} c_i$, and the normalized numerator has the residue in the formula, possibly zero. The quotient rule of Proposition 2.2 applies even if $A \cap B$ is empty. Rescaling all coefficients in the first group meeting B multiplies both residues by the same positive real, which cancels. $\square$

Equivalently, give each group the real probability proportional to its c_i . To condition, select the first group assigning positive mass to B and condition that group's law. This is a particularly transparent finite lexicographic representation of *conditional shadows*. It does not reconstruct the full surreal probability or all expected utility comparisons. For example, $(1/2 + t, 1/2 - t)$ has just the uniform leading group but prefers the first-state payoff to the second-state payoff. Higher coefficients can resolve ties even within one leading group. Conversely the conditional shadows recover the ordered groups and their coefficient ratios. For two distinct atoms i, j , the value of $C(\{i\} \mid \{i, j\})$ is 1 when $a_i < a_j$, 0 when $a_i > a_j$, and $c_i/(c_i + c_j) \in (0, 1)$ when $a_i = a_j$. Within a tied group this last number recovers c_i/c_j . Thus the group order and the normalized coefficients are exactly the information retained, but the numerical gaps between exponents and the higher coefficients are lost.

4.3 A finite compression that retains real-act comparisons

Theorem 4.3 (Signed-row lexicographic compression). *Let $p = (p_1, \dots, p_n)$ be a finite Hahn probability. There are $r \leq n$ real row vectors $u_1, \dots, u_r$ such that for every $d \in \mathbb{R}^n$ the sign of $\sum_i p_i d_i$ is the sign of the first nonzero entry in*

$$(u_1 \cdot d, \dots, u_r \cdot d), \quad (4.7)$$

and the expectation is zero exactly when all these entries are zero. The rows can be chosen from the actual coefficient rows of p in increasing exponent order.

Proof. The finite union W of the supports of the p_i is well ordered. For $\gamma \in W$ let $r_\gamma = ([t^\gamma]p_1, \dots, [t^\gamma]p_n) \in \mathbb{R}^n$. The row set is nonempty because $\sum_i p_i = 1$. Choose its earliest nonzero row. If the chosen rows do not yet span every coefficient row, the set of exponents whose rows lie outside their span is a nonempty subset of W , so has a least member. Choose that row next. The chosen rows are linearly independent, so after at most n choices they span every coefficient row. In particular, each skipped row lies in the span of chosen rows with strictly earlier exponents.

For $d \in \mathbb{R}^n$, finite distributivity gives

$$[t^\gamma] \sum_i p_i d_i = r_\gamma \cdot d.$$

If u_j is the first chosen row pairing nontrivially with d , then every earlier coefficient row pairs to zero: the chosen earlier rows do, and each skipped one is their linear combination. The exponent of u_j is therefore exactly the leading exponent of the expectation, with coefficient $u_j \cdot d$. If all chosen rows annihilate d , so do all coefficient rows, and the expectation is zero. The converse follows by coefficient extraction. $\square$

Applied to the difference of two real-valued acts, the theorem retains all their preference comparisons, including infinitesimal tie breaking. All weights have nonnegative valuation. Normalization makes the exponent-zero row a real probability vector and the sum of every later row zero. Each nonzero later row therefore has both positive and negative entries. These

selected coefficient rows need not themselves be a list of probability laws; this construction is distinct from the finite-state probability-system equivalence in [7, Section 4]. The selection argument establishes existence, not a finite search through an arbitrary infinite Hahn support.

The real-payoff hypothesis is essential. Take $p = (1/2 + t, 1/2 - t)$, whose selected rows are $(1/2, 1/2)$ and $(1, -1)$. For the surreal payoff difference $d = (-1 + t, 1)$, the first row gives $t/2 > 0$, but the actual expectation is

$$(1/2 + t)(-1 + t) + (1/2 - t) = -3t/2 + t^2 < 0.$$

Multiplication by surreal payoffs outside $\mathbb{R}$ shifts exponents, invalidating the coefficient-extraction step used in the theorem.

5 Bayesian inference at every scale

5.1 Exact posterior algebra and scale selection

For finitely many hypotheses with positive prior probabilities p_i and likelihoods $\ell_i \geq 0$, not all zero, define

$$q_i = \frac{p_i \ell_i}{Z}, \quad Z = \sum_j p_j \ell_j. \quad (5.1)$$

If the likelihood is an event probability then $\ell_i \leq 1$; if it is a density or an unnormalized score, this upper bound is unnecessary. At least one product $p_i \ell_i$ is positive, so $Z > 0$; thus every $q_i \geq 0$ and $\sum_i q_i = 1$. Multiplying all likelihoods by one positive factor multiplies Z by that factor and leaves the posterior unchanged. Bayes' formula is exact for either interpretation when the likelihood is part of a specified model.

Theorem 5.1 (Minimum-plus Bayes with coefficient tie breaking). *Suppose $p_i, \ell_i > 0$ lie in a Hahn field. Write $a_i = v(p_i)$, $b_i = v(\ell_i)$, $c_i = \text{lc}(p_i)$, $d_i = \text{lc}(\ell_i)$, and*

$$m = \min_i (a_i + b_i), \quad D = \sum_{i: a_i + b_i = m} c_i d_i. \quad (5.2)$$

Then

$$v(q_i) = a_i + b_i - m, \quad \text{lc}(q_i) = c_i d_i / D. \quad (5.3)$$

In particular, the standard posterior is supported exactly on the minimizers of $a_i + b_i$, with probabilities proportional to $c_i d_i$. Zero likelihoods may be omitted and yield zero posterior mass.

Proof. The product $p_i \ell_i$ has valuation $a_i + b_i$ and leading coefficient $c_i d_i > 0$. Equation (2.3) shows that Z has leading term Dt^m , where $D > 0$. Write $Z = Dt^m(1 + \zeta)$ with $v(\zeta) > 0$. The factor $(1 + \zeta)^{-1}$ is a unit with residue one, so division subtracts m from every numerator valuation and divides its leading coefficient by D . This proves (5.3) for every surviving index, not just the minimizers. Nonminimal terms have positive valuation and residue zero; the minimal ones have the stated positive residues, summing to one. An exactly zero likelihood has no leading coefficient and is excluded from the minimum; its posterior mass is exactly zero. $\square$

The valuation rule is minimum-plus arithmetic: multiply probabilities by adding ranks, combine alternatives by taking a minimum, and normalize by subtracting the winning rank. Leading real coefficients provide the information needed when ranks tie. It is related to ranking models [10, 9], but the exponents here belong to an ordered abelian group; their addition is not automatically ordinal addition.

Example 5.2 (Rare evidence reverses the dominant hypothesis). Take the prior (1.1) and likelihoods $(t^3, t, 1)$. The products are proportional to (t^3, t^2, t^2) , so

$$q = \left(\frac{t}{2+t}, \frac{1}{2+t}, \frac{1}{2+t} \right), \quad \text{st } q = (0, 1/2, 1/2). \quad (5.4)$$

The evidence probability is $(t^3 + 2t^2)/(1 + t + t^2) > 0$. Its standard part is zero. An analysis that first discards infinitesimal prior and likelihood terms cannot perform this update; it has erased exactly the information that the observation selects.

For finitely many nonnegative likelihood vectors $\ell^{(1)}, \dots, \ell^{(k)}$, successive updates commute provided their joint normalizer $Z_* = \sum_i p_i \prod_{j=1}^k \ell_i^{(j)}$ is positive. Some index then has positive prior and every likelihood factor positive, so every intermediate normalizer is positive in every update order. Canceling those normalizers gives $q_i = p_i \prod_j \ell_i^{(j)} / Z_*$, independent of the order. Individual updates being defined is not enough: with a positive two-state prior, likelihoods $(1, 0)$ and $(0, 1)$ each define an update, but their joint normalizer is zero and neither second update exists. Under conditional independence this product is the likelihood of the combined observations; without it, the factors must be the appropriate conditional likelihoods. For surviving hypotheses, posterior odds are prior odds times the product of likelihood ratios. No unspecified infinite product is defined by this finite identity.

5.2 How much precision does conditioning require?

For finite laws p, q define

$$\delta = \text{TV}(P, Q) = \max_A |P(A) - Q(A)| = \frac{1}{2} \sum_i |p_i - q_i|. \quad (5.5)$$

Put $h_i = p_i - q_i$ and $I_+ = \{i : h_i > 0\}$. Since $\sum_i h_i = 0$, the positive and negative totals satisfy $\sum_{I_+} h_i = -\sum_{h_i < 0} h_i = \frac{1}{2} \sum_i |h_i|$. Every event sum lies between these two signed extremes; the event I_+ attains the positive one. This proves the equality over any ordered field, including zero differences and nonregular laws.

Theorem 5.3 (Absolute and valuation-sensitive conditioning stability). *Suppose $P(B) \geq b > \delta$. Then $Q(B) > 0$ and for every A ,*

$$|P(A | B) - Q(A | B)| \leq \frac{2\delta}{b}. \quad (5.6)$$

Separately, for Hahn probabilities assume $P(B) > 0$ and put $\beta = v(P(B)) \in \Gamma$ and $\lambda \in \Gamma_{\geq 0}$. If

$$v(p_i - q_i) > \beta + \lambda \quad \text{for every atom } i, \quad (5.7)$$

then $Q(B) > 0$, it has the same leading term as $P(B)$, and

$$v(P(A | B) - Q(A | B)) > \lambda. \quad (5.8)$$

Proof. The first assumption gives $Q(B) \geq P(B) - \delta > 0$. Set $x = P(A \cap B)$, $y = Q(A \cap B)$, $d = P(B)$, and $e = Q(B)$. Then

$$\left| \frac{x}{d} - \frac{y}{e} \right| \leq \frac{|x - y|}{d} + \frac{y|e - d|}{de} \leq \frac{2\delta}{d}, \quad (5.9)$$

since $0 \leq y \leq e$. This implies (5.6). The valuation assertion does not require a separately chosen b . Write $e = d + \Delta_d$ and $y = x + \Delta_x$. Each difference is a finite sum of atom errors, so

the valuation inequality for sums gives $v(\Delta_d), v(\Delta_x) > \beta + \lambda$. Because $\lambda \geq 0$, the perturbation Δ_d cannot change the leading term of d . Thus $e > 0$ and $v(e) = v(d) = \beta$. Also $0 \leq x \leq d$ implies $v(x) \geq \beta$: a smaller valuation would make positive x dominate d . The case $x = 0$ uses $v(0) = +\infty$. The exact difference is

$$\frac{x}{d} - \frac{y}{e} = \frac{x\Delta_d - d\Delta_x}{de}.$$

Both numerator terms have valuation greater than $2\beta + \lambda$, so their difference does too; cancellation can only raise that valuation. The denominator has valuation 2β . Division proves the strict bound, including the case of an identically zero numerator. $\square$

Consequently an approximation suitable for unconditional probabilities may be unusable after rare conditioning. Input errors beyond the event scale $\beta + \lambda$ are a *sufficient* contract for preserving a conditional answer through scale λ . They are not necessary for a particular pair of laws: $(1 - 2t, t, t)$ and $(1 - 4t, 2t, 2t)$ have the same conditional law on $B = \{2, 3\}$, although their atom errors already have valuation $1 = v(P(B))$.

The strict inequality in the uniform contract cannot simply be weakened. Choose $\beta > 0$, $\lambda \geq 0$ in Γ , set $\varepsilon = \frac{1}{2}t^{\beta+\lambda}$, and take

$$p = (1 - 2t^\beta, t^\beta, t^\beta), \quad q = (1 - 2t^\beta, t^\beta + \varepsilon, t^\beta - \varepsilon).$$

Both are positive laws, and the nonzero atom errors have valuation $\beta + \lambda$. On $B = \{2, 3\}$ their first conditional coordinates differ by $t^\lambda/4$, of valuation exactly λ , not greater. At $\lambda = 0$ this is a real discrepancy. These examples isolate the precision loss without asserting necessity when numerator and denominator errors happen to cancel.

5.3 Infinite prior odds are not ordinary dogmatism

For $0 < p < 1$ and any positive likelihood ratio $R \in K$, the posterior odds are $Rp/(1-p)$. They change if and only if $R \neq 1$; a neutral ratio leaves the prior exactly unchanged. With strictly positive likelihoods both posterior coordinates remain nonzero. An exact prior endpoint, by contrast, cannot be revived by multiplying its zero weight.

An infinite prior logit cannot be brought to a limited logit by an ordinary finite number of log-likelihood increments bounded by real numbers. If L is positive infinite and $|b_j| \leq C \in \mathbb{R}$ for $j \leq n$ with ordinary finite n , then $|\sum_{j=1}^n b_j| \leq nC$ is limited. For every real M , the inequality $L > M + nC$ gives $L + \sum_j b_j > M$, so the updated logit remains positive infinite. For such a logit L' , $\exp(-L')$ is a positive infinitesimal: for every positive real a , $L' > -\log a$ implies $\exp(-L') < a$. Hence the logistic posterior $1/(1 + \exp(-L'))$ has shadow one, even when the exact posterior changes. If the increments sum to zero, the exact posterior is unchanged as well. A countervailing infinite-scale likelihood can overturn it. A statement about learning from an infinite data stream requires a chosen probability law on streams and a chosen limiting operation; it does not follow from this finite observation.

6 Surreal logits, log-scores, and softmax geometry

6.1 Binary probabilities and exact log-odds

Definition 6.1. In an exponential/logarithmic workspace E , define

$$\sigma(z) = \frac{1}{1 + \exp(-z)}, \quad \text{logit}(p) = \log \frac{p}{1-p} \quad (0 < p < 1). \quad (6.1)$$

Proposition 6.2 (Logit coordinates). *The maps $\sigma : E \rightarrow (0, 1)_E$ and $\text{logit} : (0, 1)_E \rightarrow E$ are inverse, strictly increasing bijections, and $\sigma(-z) = 1 - \sigma(z)$. For binary hypotheses with $0 < P(H) < 1$ and both likelihoods $P(D | H), P(D | H^c) > 0$, all belonging to E ,*

$$\text{logit } P(H | D) = \text{logit } P(H) + \log \frac{P(D | H)}{P(D | H^c)}. \quad (6.2)$$

Proof. For $z \in E$, positivity of $\exp(-z)$ gives $0 < \sigma(z) < 1$, and closure puts $\sigma(z)$ in E . Direct division gives $\sigma(z)/(1 - \sigma(z)) = \exp z$, hence $\text{logit}(\sigma(z)) = z$. For $0 < p < 1$, the odds $p/(1 - p)$ are positive in E ; substituting their logarithm into σ gives $\sigma(\text{logit}(p)) = p$. If $z < w$, then $\exp(-z) > \exp(-w) > 0$ and taking reciprocals of $1 + \exp(-z)$ and $1 + \exp(-w)$ proves $\sigma(z) < \sigma(w)$. Its inverse is therefore strictly increasing too. Multiplying by $\exp z$ gives $\sigma(-z) = 1 - \sigma(z)$. Both prior weights and likelihoods are positive, so the evidence mass and both posterior weights are positive. Canceling the common evidence mass in the posterior odds, then applying $\log(xy) = \log x + \log y$, proves (6.2). $\square$

Every actual surreal z , including ω or $-\omega^2$, gives a probability strictly between zero and one. The endpoints $p = 0, 1$ require formal logit symbols $-\infty, +\infty$ outside $\mathbf{No}$, not the surreals $-\omega, +\omega$. This distinction prevents an infinite confidence level from being confused with a logically impossible alternative.

For positive infinite L put $u = \exp(-L)$, which is a positive infinitesimal. Then

$$\sigma(-L) = \frac{u}{1 + u} = u - u^2 + u^3 - \dots, \quad \sigma(L) = 1 - u + u^2 - u^3 + \dots. \quad (6.3)$$

The displayed series are strong normal-form substitutions in the ambient surreals; a particular small workspace may contain the resulting rational expression without being closed under arbitrary Hahn sums. For real r and infinitesimal η , the canonical infinitesimal exponential expansion gives

$$\sigma(r + \eta) = \sigma(r) + \sigma(r)(1 - \sigma(r))\eta + O(\eta^2). \quad (6.4)$$

Here $O(\eta^2)$ means bounded by a real constant times $|\eta|^2$ in this local expansion, including zero remainder at $\eta = 0$. To see the coefficient directly, put $a = \exp(-r) > 0$ and use $\exp(-\eta) = 1 - \eta + \eta^2 R$ with limited R . Factoring $1 + a$ from $1 + a \exp(-\eta)$ leaves a principal unit. Its geometric inverse has linear term $a\eta/(1 + a)$ and a limited quadratic remainder. Multiplication by $(1 + a)^{-1}$ gives $a/(1 + a)^2 = \sigma(r)(1 - \sigma(r))$ as the linear coefficient. These are strong infinitesimal series identities, not ordinary sequence limits. For $t = \omega^{-1}$,

$$\text{logit}(t) = \log t - \log(1 - t) = -\log \omega + t + \frac{t^2}{2} + \frac{t^3}{3} + \dots. \quad (6.5)$$

Thus a probability of order ω^{-1} corresponds to a logit near $-\log \omega$, not near $-\omega$. A logit $-\omega$ specifies the much smaller probability of order $\exp(-\omega)$.

6.2 Logits do not inherit event additivity

Logits are coordinates of probabilities, not a replacement additive measure. If independent events have logits a, b , then

$$\text{logit } P(A \cap B) = a + b - \log(1 + e^a + e^b). \quad (6.6)$$

Indeed, setting $u = e^a, v = e^b$ gives $P(A \cap B) = uv/((1+u)(1+v))$, whose odds are $uv/(1+u+v)$. For disjoint events, one adds $\sigma(a) + \sigma(b)$, when this is at most one, and only then takes a logit if the result lies in $(0, 1)$. Adding $a + b$ is appropriate for independent *evidence increments in odds*, not for a disjoint union of events.

There is also a modeling distinction between a very small probability and an unnormalized score. An isolated score becomes a probability only relative to competing scores and a normalization rule. Surreal scores do not remove that requirement.

6.3 Softmax, gauge invariance, and stable normalization

For an ordinary positive integer n and scores $z_1, \dots, z_n \in E$, define

$$\text{LSE}(z) = \log \sum_{j=1}^n e^{z_j}, \quad p_i = e^{z_i - \text{LSE}(z)}. \quad (6.7)$$

All p_i are positive and sum to one. Every strictly positive finite law occurs, by taking $z_i = \log p_i$. Adding the same $c \in E$ to all scores multiplies their normalizing sum by e^c and does not change p . Conversely, equal softmax laws satisfy $z_i - w_i = \text{LSE}(z) - \text{LSE}(w)$ for every i , by taking logarithms of the equal probabilities. Thus a common additive constant is exactly the ambiguity of the score vector. For $n = 1$ the law is (1) and its logit is not an element of E . For $n \geq 2$, the one-against-the-rest logit is

$$\text{logit}(p_i) = z_i - \log \sum_{j \neq i} e^{z_j}, \quad (6.8)$$

not the raw score z_i in general.

Proposition 6.3 (Normalization and perturbation bounds). *Let $M = \max_i z_i$. Then*

$$M \leq \text{LSE}(z) \leq M + \log n. \quad (6.9)$$

If p and q are the softmax laws of z and w , and $|z_i - w_i| \leq \delta$ for every i , where $\delta \in E_{\geq 0}$, then

$$e^{-2\delta} p_i \leq q_i \leq e^{2\delta} p_i. \quad (6.10)$$

Proof. The maximum exists because $n \geq 1$ is finite. Factoring e^M from the sum leaves $\sum_i e^{z_i - M}$, whose terms lie in $(0, 1]$ and at least one of which is 1. It therefore lies in $[1, n]$; apply the increasing logarithm and add M . For perturbations, write $Z = \sum_i e^{z_i}$ and $W = \sum_i e^{w_i}$. Termwise monotonicity and finite summation give

$$e^{-\delta} \leq \frac{e^{w_i}}{e^{z_i}} \leq e^{\delta}, \quad e^{-\delta} \leq \frac{W}{Z} \leq e^{\delta}.$$

All quantities are positive. Invert the second pair of bounds and multiply it by the first to obtain $e^{-2\delta} \leq q_i/p_i \leq e^{2\delta}$. The same normalizer bounds also give $|\text{LSE}(w) - \text{LSE}(z)| \leq \delta$ after taking logarithms. $\square$

For infinitesimal δ , both $e^{2\delta} - 1$ and $1 - e^{-2\delta}$ are infinitesimal by the strong exponential expansion. Thus the displayed bounds imply $q_i/p_i - 1 \simeq 0$, even when p_i is extremely small. A finite list of infinitesimal score errors admits such a common δ by taking their maximum. This is a different guarantee from an absolute error bound on probabilities, which can be destroyed by rare conditioning. Subtracting M before exponentiating is exact algebra and avoids representing an unnecessarily large common exponential. It does not by itself provide a finite symbolic representation for every surreal score.

6.4 Hierarchical scores and lexicographic concentration

Theorem 6.4 (Finite multiscale softmax limit). *Let $m \geq 1$ be an ordinary integer and $\Omega_1, \dots, \Omega_m$ positive surreals in E with Ω_m positive infinite and Ω_{r+1}/Ω_r infinitesimal for $r < m$. For an ordinary positive number n of states suppose*

$$z_i = \sum_{r=1}^m \Omega_r a_{ir} + b_i + \eta_i, \quad a_{ir}, b_i \in \mathbb{R}, \quad \eta_i \in E, \quad \eta_i \simeq 0. \quad (6.11)$$

Let I_ be the states whose real tuple $(a_{i1}, \dots, a_{im})$ is lexicographically maximal. Then*

$$\text{st } p_i = \begin{cases} e^{b_i / \sum_{j \in I_*} e^{b_j}}, & i \in I_*, \\ 0, & i \notin I_*. \end{cases} \quad (6.12)$$

Proof. A finite nonempty set of real tuples has a lexicographic maximum, so choose $j \in I_*$. All Ω_r are positive infinite: each earlier scale exceeds the next. For $s > r$, the ratio Ω_s/Ω_r is a finite product of positive infinitesimals and is infinitesimal. If $i \notin I_*$, at the first differing coordinate r write $a_{ir} - a_{jr} = -c$ with real $c > 0$. Dividing $z_i - z_j$ by Ω_r gives $-c$ plus an infinitesimal. The later scale terms have that property by the ratio calculation; the real and infinitesimal offsets do as well because Ω_r is infinite. Hence $z_i - z_j < -c\Omega_r/2$ is negative infinite, making $e^{z_i - z_j}$ infinitesimal by comparison with the exponential of every real constant. For $i, j \in I_*$, the hierarchical terms cancel and

$$e^{z_i - z_j} = e^{b_i - b_j} e^{\eta_i - \eta_j} \simeq e^{b_i - b_j}.$$

Every term in $\sum_i e^{z_i - z_j}$ is limited. Its residue is the positive real $\sum_{i \in I_*} e^{b_i - b_j}$, with the j th term equal to 1. The standard-part quotient rule applied to this normalizer gives the asserted formula, after canceling the real factor e^{-b_j} . $\square$

The word “limit” in the theorem’s title refers to this standard-part formula for one fixed list of surreal scales; no sequence or net is taken. This supplies an exact scalar model of ordered priorities, with ordinary softmax resolving the final ties and smaller terms retaining further information. For example, $\omega^2 a_i + \omega b_i + c_i$ first maximizes a_i , then b_i , then uses c_i as a finite log-score. Here ω^2 is just the square of ω , so the exponentiation warning in Section 2 is not implicated. The standard part describes concentration; the exact law still gives positive mass to every state.

7 Finite information theory and proper scoring

7.1 Entropy and relative entropy

All sums in this section have an ordinary finite number of terms and use canonical logarithms in E . Let $n \geq 1$ be an ordinary integer. For probability vectors $p, q \in E^n$ define

$$H(p) = - \sum_{i: p_i > 0} p_i \log p_i, \quad D(p||q) = \sum_{i: p_i > 0} p_i \log \frac{p_i}{q_i}, \quad (7.1)$$

where the relative entropy is defined in E if $q_i > 0$ whenever $p_i > 0$. If that condition fails, it is useful to assign a formal value $+\infty$; that value is not a surreal and is not used in field manipulations. The convention $0 \log 0 = 0$ refers to entropy, not to an evaluation of $\log 0$.

Theorem 7.1 (Finite Gibbs inequality and entropy bounds). *Under the support condition in (7.1),*

$$D(p\|q) \geq 0, \quad D(p\|q) = 0 \iff p = q. \quad (7.2)$$

For n states,

$$0 \leq H(p) \leq \log n, \quad \text{st } H(p) = H(\text{st } p). \quad (7.3)$$

Proof. Let $S = \{i : p_i > 0\}$. Apply $-\log x \geq 1 - x$ to $x = q_i/p_i$ and multiply by $p_i > 0$. Summing gives

$$D(p\|q) \geq \sum_{i \in S} (p_i - q_i) = 1 - \sum_{i \in S} q_i \geq 0. \quad (7.4)$$

More explicitly, the exact decomposition is

$$D(p\|q) = \sum_{i \in S} p_i \left(-\log \frac{q_i}{p_i} - 1 + \frac{q_i}{p_i} \right) + \sum_{i \notin S} q_i.$$

Every summand is nonnegative. A finite sum of nonnegative elements of an ordered field is zero exactly when each summand is zero; the strict scalar logarithm inequality then forces $q_i = p_i$ on S and zero mass outside S . The converse follows by $\log 1 = 0$. Each $-p_i \log p_i$ is nonnegative. Applying Gibbs inequality to the uniform law gives $D(p\|(1/n, \dots, 1/n)) = \log n - H(p) \geq 0$.

For the standard-part identity, if $\text{st } p_i = r_i > 0$, write $p_i = r_i(1 + \xi_i)$ with infinitesimal ξ_i . Then $\log p_i = \log r_i + \log(1 + \xi_i) \simeq \log r_i$, so the ring law gives $\text{st}(-p_i \log p_i) = -r_i \log r_i$. If $p_i > 0$ is infinitesimal, apply the transferred real-exponential inequality $\log x \leq \sqrt{x}$ for $x \geq 4$ to $x = 1/p_i$. Here the positive root belongs to E , since $\sqrt{x} = \exp(\frac{1}{2} \log x)$. This gives

$$0 \leq p_i \log(1/p_i) \leq \sqrt{p_i} \simeq 0.$$

The last infinitesimality follows from $p_i < a^2$ for every positive real a . Zero coordinates contribute the specified zero term. Taking standard parts of the finite sum now proves the formula. This argument does not take the standard part of $\log p_i$ when that logarithm is infinite. $\square$

An entropy on a fixed finite alphabet is always limited. It nevertheless retains infinitesimal information discarded by its standard part. For a rare Bernoulli success of probability t ,

$$H(t, 1 - t) = t \log(1/t) + t - \frac{t^2}{2} + O(t^3). \quad (7.5)$$

Indeed $-\log(1 - t) = \sum_{k \geq 1} t^k/k$ strongly; multiplication by $1 - t$ gives $t - \sum_{k \geq 2} t^k/(k(k - 1))$, whose tail after degree two is t^3 times a limited element. Here the remainder notation again means a real bound times t^3 . The log factor is essential: entropy need not remain in the original pure Laurent field $\mathbb{R}((t))$. It belongs to the exponential/logarithmic workspace, where the displayed products are meaningful. This is one reason not to assume that a single Hahn field is automatically closed under all information-theoretic operations.

7.2 Conditional information and data processing

For a finite joint law write $p_{ij} = p_j^Y p_i^{X|j}$ on every cell with $p_j^Y > 0$. Expanding $\log(p_j^Y p_i^{X|j})$ and summing yields

$$H(X, Y) = H(Y) + \sum_j p_j^Y H(X | Y = j), \quad (7.6)$$

where zero-mass cells contribute zero. The same decomposition for two joint laws satisfying the relative-entropy support condition gives

$$D(p_{XY} \| q_{XY}) = D(p_Y \| q_Y) + \sum_{j: p_j^Y > 0} p_j^Y D(p_{X|j} \| q_{X|j}). \quad (7.7)$$

If $p_j^Y > 0$, some $p_{ij} > 0$, so the joint support condition gives $q_{ij} > 0$ and hence $q_j^Y > 0$. Both conditional laws are therefore defined, and their support condition also holds. On a cell with $p_{ij} > 0$, factor the joint ratio as $(p_j^Y/q_j^Y)(p_i^{X|j}/q_i^{X|j})$ and split its logarithm. Summing the marginal term over i contributes $p_j^Y \log(p_j^Y/q_j^Y)$, and the remaining terms give the conditional divergence. Columns with $p_j^Y = 0$ contribute nothing; their conditional laws need not be specified. All regrouping is finite.

Theorem 7.2 (Finite data processing). *Let $T_{ij} \geq 0$ be a finite stochastic matrix, $\sum_j T_{ij} = 1$. For p, q satisfying the support condition,*

$$D(pT \| qT) \leq D(p \| q). \quad (7.8)$$

Equality holds exactly when, for each output j with $(pT)_j > 0$, the two input conditional laws given j coincide.

Proof. Form the joint laws $P_{ij} = p_i T_{ij}$ and $Q_{ij} = q_i T_{ij}$. If $P_{ij} > 0$, then $p_i > 0$ and $T_{ij} > 0$, so $q_i > 0$ and $Q_{ij} > 0$. Thus the joint support condition holds, and on this cell $P_{ij}/Q_{ij} = p_i/q_i$. Summing over j and using $\sum_j T_{ij} = 1$ gives $D(P \| Q) = D(p \| q)$. The output marginals are pT, qT , with the inherited support condition. The chain rule now gives the exact nonnegative difference

$$D(p \| q) - D(pT \| qT) = \sum_{j: (pT)_j > 0} (pT)_j D(P_{X|j} \| Q_{X|j}).$$

Its weights are positive even when infinitesimal. A finite sum of such nonnegative terms vanishes precisely when each conditional divergence vanishes, which is equivalent to equality of the entire two conditional vectors by Theorem 7.1. No condition is imposed at outputs with zero pT mass; their terms are absent. $\square$

Taking $q_{ij} = p_i^X p_j^Y$ gives the nonnegative mutual information $I(X; Y) = D(p_{XY} \| p_X p_Y)$ and the usual finite identities $I(X; Y) = H(X) + H(Y) - H(X, Y)$ and $I(X; Y) = H(X) - H(X | Y)$. Independence is equivalent to zero mutual information. A process-level entropy rate would involve a limit in the horizon and is not supplied by these finite formulas.

7.3 Identical real shadows can hide infinite divergence

Example 7.3 (An information scale invisible to standard probability). Let $t = \omega^{-1}$ and $u = \exp(-1/t^2)$. Consider

$$p = (t, 1 - t), \quad q = (u, 1 - u). \quad (7.9)$$

The inequality $\exp x \geq 1 + x$ at $x = 1/t^2$ gives $0 < u \leq t^2/(1 + t^2)$, so u is positive infinitesimal. Both standard parts equal $(0, 1)$, but

$$D(p \| q) = \frac{1}{t} + t \log t + (1 - t)(\log(1 - t) - \log(1 - u)), \quad (7.10)$$

$$\text{st}(tD(p \| q)) = 1. \quad (7.11)$$

Thus their relative entropy is positive infinite. In fact the final term in (7.10) is infinitesimal because both $\log(1 - t)$ and $\log(1 - u)$ are infinitesimal and $1 - t$ is limited. The entropy estimate above gives $t \log t \simeq 0$. Thus $D(p \| q) = 1/t + \varepsilon$ for an infinitesimal ε , which proves both positive infinitude and the displayed normalized residue.

There is no conflict with (7.3). Entropy uses each law's own probabilities and is bounded on a fixed alphabet. Relative entropy can compare a rare event against a vastly smaller prediction. Taking standard parts before forming the logarithmic ratio loses this penalty. Standard part therefore commutes with finite entropy but does not in general commute with relative entropy, even when the latter has a well-defined surreal value.

7.4 Strictly proper scores survive

For an outcome i , logarithmic loss is $-\log q_i$. If all predicted probabilities are positive, its expected excess over truthful prediction is exactly

$$\mathbb{E}_p[-\log q_i] - H(p) = D(p\|q). \quad (7.12)$$

If the true law p is strictly positive, the loss over strictly positive predictions is therefore uniquely minimized by $q = p$. If p has a zero coordinate, p is outside that prediction domain and no minimizer in it exists. Indeed let $S = \{i : p_i > 0\}$ and $a = \sum_{i \in S} q_i < 1$. Choose $a < c < 1$, multiply the predicted weights on S by c/a and those outside S by $(1-c)/(1-a)$. The new prediction remains strictly positive and normalized, while its expected loss decreases by $\log(c/a) > 0$.

The lower bound $H(p)$ is still the infimum. For $q^{(\varepsilon)} = (1-\varepsilon)p + \varepsilon(1/n, \dots, 1/n)$ with $0 < \varepsilon < 1$,

$$0 \leq D(p\|q^{(\varepsilon)}) \leq -\log(1-\varepsilon),$$

since $q_i^{(\varepsilon)} \geq (1-\varepsilon)p_i$ on S . Given any $\delta > 0$ in E , take $\varepsilon = 1 - \exp(-\delta/2)$ to make the excess at most $\delta/2$. This is an order-infimum argument inside E , with no convergence convention or evaluation of $\log 0$. The Brier loss $S(q, i) = \sum_j (q_j - \mathbf{1}_{j=i})^2$ satisfies the entirely algebraic identity

$$\mathbb{E}_p S(q, i) - \mathbb{E}_p S(p, i) = \sum_j (q_j - p_j)^2. \quad (7.13)$$

For completeness, $\mathbb{E}_p S(q, i) = \sum_j q_j^2 - 2 \sum_j p_j q_j + 1$; subtracting the same expression at $q = p$ gives the displayed sum of squares. It is nonnegative, with equality only when every $q_j = p_j$. Thus Brier loss is strictly proper on the whole finite simplex over any ordered field, including boundary laws. These scores distinguish infinitesimal prediction errors, though their relative penalties can occur at different scales.

For a binary soft target $0 < p < 1$ and logit z , the expected logarithmic loss is

$$\Phi(z; p) = \log(1 + e^z) - pz. \quad (7.14)$$

Indeed $-\log \sigma(z) = \log(1+e^z) - z$ and $-\log(1-\sigma(z)) = \log(1+e^z)$, giving the displayed expected loss. Since $q = \sigma(z)$ parametrizes the strictly positive binary simplex, $\Phi(z; p) - H(p, 1-p)$ is exactly the corresponding relative entropy. Its unique minimizer is therefore $z = \text{logit}(p)$. This conclusion uses an explicit identity and positivity, not an unproved extreme-value theorem on a surreal parameter space.

8 Gibbs laws, decision theory, and finite learning models

8.1 A finite variational principle without compactness

Theorem 8.1 (Surreal finite Gibbs principle). *Let $n \geq 1$ be an ordinary integer, let $E_1, \dots, E_n \in E$ be energies, and let $\tau > 0$ belong to E . Set*

$$Z = \sum_i e^{-E_i/\tau}, \quad q_i = Z^{-1} e^{-E_i/\tau}. \quad (8.1)$$

For every probability vector $p \in E^n$,

$$\sum_i p_i E_i - \tau H(p) = -\tau \log Z + \tau D(p||q). \quad (8.2)$$

Hence q is the unique minimizer of the left-hand side on the finite simplex, even when some energies or $1/\tau$ are infinite.

Proof. Each summand defining Z is positive and lies in E ; thus $Z > 0$ and q is a strictly positive probability vector in E^n . Consequently $D(p||q)$ is defined for every p on the entire simplex, even when some $p_i = 0$. On its positive support, substitute $\log q_i = -E_i/\tau - \log Z$ to obtain

$$D(p||q) = -H(p) + \frac{1}{\tau} \sum_i p_i E_i + \log Z,$$

using $\sum_i p_i = 1$. Multiplication by the positive τ gives the identity, and Gibbs inequality makes its excess term nonnegative, with equality exactly at $p = q$. A positive infinitesimal τ preserves strict positivity of a nonzero excess just as any other positive field element does. At $n = 1$, $q = (1)$ and the identity reduces to $E_1 = E_1$. $\square$

No compactness or Dedekind completeness of the simplex was invoked. The minimizer was written down, and the identity proved optimality. For a positive infinitesimal temperature and real energies, let $E_* = \min_i E_i$ and let $G = \{i : E_i = E_*\}$. Factor $e^{-E_*/\tau}$ out of Z . Every ground-state term is 1; every excited term is $e^{-(E_i - E_*)/\tau}$, infinitesimal because its positive real gap divided by τ is infinite. The normalized denominator therefore has residue $|G|$, giving shadow $1/|G|$ on G and zero elsewhere. The gap relative to temperature matters: for energies $(0, \tau)$ the law is exactly $(1, e^{-1})/(1 + e^{-1})$, even when τ is infinitesimal, so both shadows are positive. Several ordered energy scales can be treated by Theorem 6.4. This does not construct a thermodynamic limit, an infinite-volume Gibbs measure, or a partition function over an arbitrary infinite state space.

8.2 Expected utility and the relevance of rare events

For a nonempty finite collection of acts on finitely many states, expected utility is an ordered-field calculation. Comparing the finite nonempty list of expected utilities successively produces a maximum; no supremum or compactness argument is required. No Archimedean continuity of preferences is forced: a positive infinitesimal preference can be weaker than every positive real preference and still be strict. The compression in Theorem 4.3 explains how real-valued acts can detect successive signed coefficient layers.

There is an additional reason not to delete rare events when payoffs are allowed to be large. An event of probability t with loss $1/t$ has expected loss exactly one. Replacing its probability by zero before multiplication gives the wrong answer. By contrast, for finitely many *limited* payoffs X_i , standard part obeys

$$\text{st } \mathbb{E}_p X = \sum_i (\text{st } p_i)(\text{st } X_i), \quad (8.3)$$

Every product $p_i X_i$ and their finite sum are limited, so the ring property of standard part applies. This explains precisely why the qualification on the payoffs is indispensable. In the rare-loss example $X = \mathbf{1}_A/t$ with $P(A) = t$, the payoff $1/t$ has no real standard part, although the expectation is the ordinary real number 1. A normative application must decide which utilities and trades it permits; surreal arithmetic does not itself settle that decision.

8.3 Infinitesimal smoothing

Suppose a nonempty finite alphabet has ordinary counts $N_i \in \mathbb{N}$ with $N = \sum_i N_i > 0$. Choose $a_i \in \mathbb{R}_{>0}$ and a positive infinitesimal t in the scalar field. The law

$$p_i = \frac{N_i + ta_i}{N + t \sum_j a_j} \quad (8.4)$$

is regular: its numerator and denominator are positive, and summing the numerators gives the denominator. The denominator has residue N , so a seen category has standard probability N_i/N . For an unseen category,

$$p_i = t \frac{a_i}{N + t \sum_j a_j}, \quad \text{st}(p_i/t) = a_i/N.$$

In Hahn coordinates this means $v(p_i) = v(t)$ and $\text{lc}(p_i) = (a_i/N) \text{lc}(t)$; the leading coefficient is literally a_i/N when t is the unit-coefficient monomial ω^{-1} . Any single nonnegative likelihood vector that is not identically zero has a positive normalizer for this regular prior. Its posterior is positive exactly on the positive-likelihood categories. A sequence of updates needs the positive joint normalizer from Section 5; smoothing cannot make incompatible zero likelihoods compatible. The constants a_i and the scale t are modeling choices, not consequences of the data. Distinct choices can give the same empirical shadow and different answers after rare conditioning.

For $N = 0$, all nonnegative counts vanish. Canceling the nonzero t in (8.4) gives the ordinary prior $a_i/\sum_j a_j$; this case should be handled explicitly rather than described as an infinitesimal perturbation of absent empirical frequencies.

8.4 Surreals do not automatically cure separation

Proposition 8.2 (No attained logistic optimum under strict separation). *For a nonempty finite collection of feature vectors $x_i \in E^d$ and labels $y_i \in \{-1, 1\}$, define*

$$L(\theta) = \sum_i \log(1 + \exp(-y_i x_i \cdot \theta)). \quad (8.5)$$

If there exists $v \in E^d$ with $y_i x_i \cdot v > 0$ for every i , then L has no minimizer in E^d .

Proof. Fix any $\theta \in E^d$ and any $s > 0$ in E . Each signed margin increases by $s(y_i x_i \cdot v) > 0$ at $\theta + sv$. Its negative therefore decreases, and strict monotonicity of both $\exp$ and $\log$ gives

$$\log(1 + e^{-y_i x_i \cdot (\theta + sv)}) < \log(1 + e^{-y_i x_i \cdot \theta}).$$

The data set is nonempty, so summing its finitely many strict inequalities strictly lowers L . The new parameter still belongs to E^d , even when θ or v has infinite coordinates. Thus every candidate is improved; no limiting point, derivative or continuity theorem is used. $\square$

An infinite parameter is not a terminal point at infinity: it can still be moved in the separating direction. Infinitesimal positive losses are not zero. Without strict separation a minimizer can exist. For example, two observations with the same scalar feature $x = 1$ and opposite labels have loss $\log((1 + e^{-\theta})(1 + e^{\theta}))$. Writing $u = e^{\theta} > 0$, one has $u + u^{-1} - 2 = (u - 1)^2/u \geq 0$, with equality only at $\theta = 0$. Their loss therefore has its unique minimum $\log 4$ at zero. Soft targets with independent logits have the explicit optimum described in (7.14); that fact does not establish existence for every constrained regression or regularization problem. Such problems need their own hypotheses and proofs.

9 Finite stochastic processes and the time-scale issue

9.1 Finite chains and finite paths

Let the finite nonempty state set be S , let T be a stochastic matrix over K indexed by S , and let π be an initial law. For each ordinary integer horizon $N \geq 0$, use all subsets of S^{N+1} as events and define the path weights

$$P(i_0, \dots, i_N) = \pi_{i_0} \prod_{k=1}^N T_{i_{k-1}, i_k} \quad (9.1)$$

using the empty product 1 at $N = 0$. They are nonnegative. For any fixed prefix, summing its extensions multiplies its weight by $\sum_{i_N \in S} T_{i_{N-1}, i_N} = 1$. Repeatedly eliminate terminal coordinates to prove normalization and consistency with every shorter horizon, ending with $\sum_{i_0} \pi_{i_0} = 1$. All initial and transition entries lie in $[0, 1]$. Their standard parts are therefore defined, and the row sums remain 1. The ring property of standard part applied to each path product and each finite event sum gives exactly the real chain with initial law $\text{st } \pi$ and transition matrix $\text{st } T$. This consistency alone is not a theorem about a probability measure on infinite paths; Section 12 supplies concrete obstructions to one proposed countable-additivity rule.

The repository's analysis of finite Hahn Markov generators and their resolvents concerns another important part of multiscale stochastic structure [15]. A stochastic matrix at finitely many steps, a continuous-time semigroup, and a path measure are different objects. This article only asserts a path construction when the relevant finite or infinite probability framework has been specified.

9.2 Finite martingales and stopping

On a finite probability space with a filtration $\mathcal{F}_0 \subseteq \dots \subseteq \mathcal{F}_N$ of event algebras, a martingale $(M_k)_{k=0}^N$ is K -valued, has M_k measurable with respect to $\mathcal{F}_k$, and satisfies $\mathbb{E}[M_k \mid \mathcal{F}_{k-1}] = M_{k-1}$ outside a null event for $1 \leq k \leq N$. The conditional expectation is the one already constructed by finite partitions; null-atom versions do not affect scalar expectations. A stopping time $T : \Omega \rightarrow \{0, \dots, N\}$ satisfies $\{T \leq k\} \in \mathcal{F}_k$ for every k . Here T denotes the random time, not the transition matrix of the preceding subsection.

Proposition 9.1 (Bounded finite-horizon optional stopping). *If T is a stopping time taking values in $\{0, \dots, N\}$, then $\mathbb{E}M_T = \mathbb{E}M_0$.*

Proof. The pathwise identity

$$M_T = M_0 + \sum_{k=1}^N \mathbf{1}_{\{T \geq k\}} (M_k - M_{k-1}) \quad (9.2)$$

is a pointwise telescoping sum: on $\{T = j\}$ exactly the terms $1 \leq k \leq j$ occur, including the empty sum when $j = 0$. The stopping-time property gives $\{T \geq k\} = \Omega \setminus \{T \leq k-1\} \in \mathcal{F}_{k-1}$. The pull-out rule and the martingale identity therefore give

$$\mathbb{E}[\mathbf{1}_{\{T \geq k\}} (M_k - M_{k-1})] = \mathbb{E}[\mathbf{1}_{\{T \geq k\}} \mathbb{E}[M_k - M_{k-1} \mid \mathcal{F}_{k-1}]] = 0.$$

Any null-atom version contributes zero. Taking expectations of the finite sum proves the claim, including $N = 0$. $\square$

No boundedness by a real constant of M_k is needed on this finite space. Conversely, this proof does not justify an unbounded stopping time or a limit of stopped martingales. Those assertions require additional integration and convergence results. The stopping-time hypothesis is also essential: on two equiprobable states take $M_0 = 0$, $M_1 \in \{-1, 1\}$, trivial $\mathcal{F}_0$ and full $\mathcal{F}_1$. Choosing $T = 1$ exactly when $M_1 = 1$ gives $\mathbb{E}M_T = 1/2$, but is not a stopping time because $\{T \leq 0\}$ is not in $\mathcal{F}_0$. It selects whether to stop using information that was unavailable at time zero.

9.3 Rare transitions and sample size

Consider a state that changes irreversibly with probability $t > 0$ infinitesimal at each step while it remains in its initial state. At a fixed ordinary horizon $n \geq 0$ its hitting probability is

$$1 - (1 - t)^n \leq nt. \quad (9.3)$$

The no-change path has probability $(1 - t)^n$, so its complement has probability $1 - (1 - t)^n$. For the bound, start with equality at $n = 0$ and use

$$1 - (1 - t)^{n+1} = 1 - (1 - t)^n + t(1 - t)^n \leq nt + t.$$

Here $0 \leq (1 - t)^n \leq 1$. Thus the hitting probability is infinitesimal at every ordinary horizon; it is strictly positive when $n \geq 1$ and zero when $n = 0$. Its standard part is always zero. Writing “take $n = 1/t$ ” is not a valid use of this formula unless a meaning for a nonstandard or surreal-indexed number of trials has first been supplied. A scalar in **No** is not automatically a finite set cardinality, a path length, or an induction index.

For an ordinary positive integer n , let $X_1, \dots, X_n$ be independent Bernoulli variables with parameter $p \in [0, 1]_K$ and put $\bar{X}_n = n^{-1} \sum_{k=1}^n X_k$. Finite variance algebra gives

$$\mathbb{E}\bar{X}_n = p, \quad \text{Var}(\bar{X}_n) = \frac{p(1-p)}{n}, \quad P(|\bar{X}_n - p| \geq a) \leq \frac{p(1-p)}{na^2}. \quad (9.4)$$

Indeed $X_k^2 = X_k$ gives $\text{Var}(X_k) = p(1-p)$, and independence gives $\mathbb{E}[X_j X_k] = p^2$ for $j \neq k$, so all cross covariances vanish. Expanding the square of the finite sum yields variance $np(1-p)$; dividing by n^2 and applying Chebyshev proves the display. The inequalities hold for surreal $a > 0$, including the deterministic cases $p = 0, 1$. Their interpretation depends on scale. For fixed positive real a and a real shadow, they have the usual ordinary limiting implications in a compatible classical model. For an infinitesimal resolution a , increasing an ordinary integer n need not make the right-hand side small. At $p = a = t$, for example, the bound $(1 - t)/(nt)$ is infinite for every ordinary $n \geq 1$. In fact $P(|\bar{X}_n - t| \geq t) = 1$: the empirical mean is either zero, with error exactly t , or at least $1/n > 2t$, with error greater than t . This is an exact finite experiment at each n , not a claim that an infinite path law has already been constructed. There is no unconditional law of large numbers in the full fine surreal topology hidden in this finite calculation.

10 Why infinite addition is an additional axiom

10.1 The fine topology is too strong for ordinary probabilistic limits

Proposition 10.1 (Set-indexed fine convergence is eventually constant). *If a net $(x_i)_{i \in I}$ with a nonempty set-sized directed index set converges to x in the full ordered-field topology of **No**, then $x_i = x$ eventually.*

Proof. The nonzero distances form a set $D = \{|x_i - x| : x_i \neq x\}$ of positive surreals. If D is empty the claim is immediate. Otherwise the surreal cut property gives a positive δ smaller than every member of D , for example a number between $\{0\}$ and D . Convergence makes $|x_i - x| < \delta$ eventually. By construction this forces $x_i = x$. Conversely, a net eventually equal to x converges to x , so this describes all such convergent nets. $\square$

The statement concerns the full fine topology and set-sized indices. It does not concern an intrinsic topology on a fixed subfield: the δ chosen in the proof may lie outside that subfield. For a universe-sized formal carrier, the permitted index sets must be small relative to that carrier, as in the foundations theorem; quantifying over all subsets of the carrier would remove the crucial size hypothesis. This does not make the full fine topology discrete: the ball of radius $r > 0$ about x contains the distinct point $x + r/2$. Nevertheless, even in any ordered subfield containing a positive infinitesimal t , the ordinary real partial sums of $\sum_{n \geq 1} 2^{-n}$ do not converge to one in the field's order topology. Their real tails 2^{-N} exceed t for every ordinary N . Coefficientwise real convergence and intrinsic valuation convergence are different again. These distinctions agree with the repository's foundations discussion [14], and Proposition 10.1 is clause (2) of a theorem of that report (Section 1.5).

Thus ordinary countable additivity cannot be imported by writing an unqualified order-topological limit of finite partial sums. A successful infinite theory must choose a different addition convention, a different topology, or a standard-part procedure.

10.2 What strong Hahn addition does and does not allow

Work in one Hahn field $\mathbb{R}((t^\Gamma))$, with Γ a set-sized ordered abelian group. A strong Hahn probability is a nonnegative event map on a sigma-algebra, with $P(\emptyset) = 0$ and $P(\Omega) = 1$, such that for every disjoint sequence (A_n) the family $(P(A_n))$ is strongly Hahn summable and

$$P\left(\bigcup_n A_n\right) = \sum_n^H P(A_n). \quad (10.1)$$

Here strong summability requires a well-ordered union of supports and only finitely many nonzero contributions at each exponent. In particular, finite additivity follows by appending empty events to a finite disjoint list. The ordinary series $\sum_{n \geq 1} 2^{-n}$ is not strong: every summand contributes to exponent zero. An infinite collection of equal positive infinitesimal singleton masses is not strong either, since the leading monomial repeats infinitely often. These are failures of a specified support rule, not contradictions in the ordered field or in finite additivity.

Example 10.2 (A regular strong Hahn probability on a countably infinite space). On $\mathbb{N}_{\geq 1}$ with all subsets measurable, take $t = \omega^{-1}$ and

$$w_n = (1 - t)t^{n-1}, \quad P(A) = \sum_{n \in A}^H w_n. \quad (10.2)$$

The supports lie in $\mathbb{N}$, a well-ordered set, and $w_n = t^{n-1} - t^n$ contributes only at two consecutive exponents. Every exponent therefore occurs in at most two weights, proving strong summability of all subfamilies. With $\mathbf{1}_A(0) = 0$, the coefficient of t^k in $P(A)$ is

$$\mathbf{1}_A(k+1) - \mathbf{1}_A(k) \quad (k \geq 0).$$

For the whole space this is one at $k = 0$ and zero thereafter, so $P(\mathbb{N}_{\geq 1}) = 1$. For nonempty A , its least element m makes all coefficients below $m - 1$ zero and the coefficient at $m - 1$ one; hence $P(A) > 0$. In a disjoint sequence of events, each of the two indices $k, k + 1$ can belong to at most one event. Thus at most two event probabilities contribute at k , and their coefficients sum

to that of the union. This proves strong countable additivity directly. The constant coefficient is $\mathbf{1}_A(1)$, so the standard part is the point mass at 1. More explicitly, $P(\{n > N\}) = t^N > 0$ for every ordinary $N \geq 0$; the first N weights sum to $1 - t^N$, not to one.

Strong probabilities on countably separated spaces are much more rigid than arbitrary finite-additive or coefficientwise laws. The repository's classification gives finitely atomic coefficient measures and a finitely supported standard part in that class [13] (Section 1.5, item (b)). We use the explicit example and the elementary support obstructions above without requiring the full classification proof. The real Lebesgue law, embedded as exponent zero, is not strong because disjoint positive-measure intervals make that exponent recur infinitely often.

10.3 Coefficientwise countable additivity

Definition 10.3. A coefficientwise Hahn measure on $(\Omega, \mathcal{A})$ is specified by a set-sized well-ordered set $W \subseteq \Gamma$ and ordinary countably additive signed measures μ_γ of finite total variation on $\mathcal{A}$, via

$$P(A) = \sum_{\gamma \in W} \mu_\gamma(A) t^\gamma. \quad (10.3)$$

It is a probability when these Hahn values are nonnegative on every A and $P(\Omega) = 1$. Countable additivity means that each coefficient uses the ordinary absolutely convergent scalar sum on a disjoint sequence:

$$\mu_\gamma\left(\bigcup_n A_n\right) = \sum_n \mu_\gamma(A_n), \quad \sum_n |\mu_\gamma(A_n)| \leq |\mu_\gamma|(\Omega) < \infty.$$

Here $|\mu_\gamma|$ denotes ordinary total variation. The one common set W controls all events; a different support bound for each event would not be this definition.

The definition assumes a sigma-algebra and ordinary countably additive finite signed measures. It is strictly more permissive than the strong rule in familiar separated spaces: real probability measures, diffuse or atomic, are included as single exponent-zero coefficients. Higher coefficient measures may be signed even when the reassembled Hahn probability is positive. Positivity of every cylinder in a product or of every finite test does not automatically imply positivity on all measurable sets; the repository treats this issue explicitly [13]. In the constructions below, positivity is proved directly rather than inferred from finite tests alone.

11 Diffuse hierarchical probabilities and integration

11.1 A positive construction with arbitrarily many ordered layers

Theorem 11.1 (Normalized hierarchy of real probability laws). *Let $(\mu_\alpha)_{\alpha < \kappa}$ be a nonempty set-sized well-ordered family of ordinary real probability measures on one measurable space. Choose strictly increasing exponents $\gamma_\alpha \in \Gamma$ with $\gamma_0 = 0$. Define*

$$Z = \sum_{\alpha < \kappa}^H t^{\gamma_\alpha}, \quad P(A) = Z^{-1} \sum_{\alpha < \kappa}^H t^{\gamma_\alpha} \mu_\alpha(A). \quad (11.1)$$

Then P is a coefficientwise Hahn probability. An event B has positive mass exactly when some $\mu_\alpha(B) > 0$. If $\alpha(B)$ is the first such index, then

$$\text{st } P(A \mid B) = \frac{\mu_{\alpha(B)}(A \cap B)}{\mu_{\alpha(B)}(B)}. \quad (11.2)$$

In particular $\text{st } P = \mu_0$.

Proof. Put $W_0 = \{\gamma_\alpha : \alpha < \kappa\}$. The strictly increasing image of the ordinal κ is well ordered, and each exponent receives one monomial contribution. Thus both displayed sums exist, with all numerator supports contained in W_0 . Write $Z = 1 + U$, where U has strictly positive support. The positive-support Neumann lemma makes $\sum_{n \geq 0}^H (-U)^n$ a Hahn series and the inverse of Z ; when $\kappa = 1$, $U = 0$ and this is just one. In particular, Z and Z^{-1} have leading coefficient one at exponent zero.

If every $\mu_\alpha(A)$ is zero, the numerator is zero. Otherwise its first nonzero coefficient is positive, since every component is a positive real measure; multiplication by $Z^{-1} > 0$ preserves its sign. The numerator for Ω is Z , giving normalization. Finite additivity holds in each coefficient, so complements also give $P(A) \leq 1$.

For the full coefficientwise definition, write $Z^{-1} = \sum_{\delta \in V} c_\delta t^\delta$, with V well ordered and extend c_δ by zero off V . The Hahn product lemma makes $W = V + W_0$ well ordered and gives only finitely many pairs (δ, γ_α) at each exponent η . Consequently

$$[t^\eta]P(A) = \sum_{\alpha: \eta - \gamma_\alpha \in V} c_{\eta - \gamma_\alpha} \mu_\alpha(A).$$

This is a finite real linear combination of ordinary probability measures, with total variation at most $\sum_{\alpha: \eta - \gamma_\alpha \in V} |c_{\eta - \gamma_\alpha}|$. The support bound W is independent of A . Each coefficient is therefore an ordinary finite signed measure, proving coefficientwise countable additivity.

For $P(B) > 0$, put $b = \alpha(B)$. Cancel Z^{-1} in the conditional ratio and divide both numerator and denominator by t^b . All coefficients before b in the numerator vanish because $0 \leq \mu_\alpha(A \cap B) \leq \mu_\alpha(B) = 0$ there. The denominator now has positive real residue $\mu_b(B)$; the numerator has residue $\mu_b(A \cap B)$, possibly zero. Both are limited, so standard-part division proves (11.2). Taking $B = \Omega$, whose first active component is zero, gives $\text{st } P = \mu_0$. $\square$

This construction is not restricted to finite or countable hierarchies. Its existence uses set-sized well-ordered support, not a real-convergent sum of the coefficients of Z . In particular, one must not evaluate $t = 1$ in (11.1) and regard the generally divergent result as an ordinary mixture. The law is regular precisely when every nonempty measurable event has positive mass for at least one component. An arbitrary hierarchy need not satisfy that coverage condition.

Example 11.2 (Diffuse probability with exceptional point events). On the Borel subsets of $[0, 1]$ let λ be Lebesgue probability and put

$$P(A) = \frac{\lambda(A) + t\delta_0(A) + t^2\delta_1(A)}{1 + t + t^2}. \quad (11.3)$$

Its standard part is exactly Lebesgue probability, but both endpoint singletons have positive infinitesimal mass. Conditioning on $\{0, 1\}$ gives masses $1/(1+t)$ and $t/(1+t)$, with conditional shadow concentrated at zero. Other singletons retain exact probability zero. Thus this is not an all-subsets regular probability, and it does not solve conditioning on every exact-null event.

Higher coefficients of this normalized law are signed because division by $1 + t + t^2$ introduces corrections: its coefficient at t is $\delta_0 - \lambda$, negative on any positive-Lebesgue-measure event that omits zero. Those signed coefficients are compatible with positivity of every event, which was proved before expanding the reciprocal. This is a useful construction pattern: prove positivity of the normalized expression rather than demand positivity of every coefficient.

11.2 A controlled expectation for bounded real observables

For the hierarchy in Theorem 11.1 and a bounded measurable *real-valued* function f , define

$$\mathbb{E}_P f = Z^{-1} \sum_{\alpha < \kappa}^H t^{\gamma_\alpha} \int f d\mu_\alpha. \quad (11.4)$$

Every component integral is a finite real number, and the numerator support lies in W_0 from the proof above, so this defines a Hahn value. Ordinary linearity and finite Hahn addition give real linearity, and indicators recover $P(A)$. If $f \geq 0$, all numerator coefficients are nonnegative, proving positivity after division by $Z > 0$. Constants are preserved: $\mathbb{E}_P c = c$. Applying positivity to $M \pm f$ whenever $|f| \leq M$ gives $|\mathbb{E}_P f| \leq M$, in particular $|\mathbb{E}_P f| \leq \|f\|_\infty$, where this norm is the pointwise supremum bound, not an essential supremum for μ_0 alone. The constant coefficient gives $\text{st } \mathbb{E}_P f = \int f d\mu_0$. For example, the endpoint indicator in the preceding example has μ_0 -essential supremum zero but positive surreal expectation.

Let f_n, f be bounded measurable real functions with one uniform real bound, and suppose $f_n \rightarrow f$ almost everywhere for each component μ_α . The exceptional null set may depend on α ; no uncountable intersection of full-measure sets is needed. Ordinary dominated convergence gives convergence of each component integral. The coefficient formula in the proof of Theorem 11.1 is a finite linear combination of those integrals at every fixed exponent. Thus $\mathbb{E}_P f_n$ converges *coefficientwise in the real coefficient topology* to $\mathbb{E}_P f$. This is a precise dominated-convergence statement, but it is not fine or ordered-field convergence. For example, the constant functions $f_n = 1/n$ have expectations $1/n$, which converge coefficientwise to zero but stay above any fixed positive infinitesimal. Their errors all have valuation zero, so they do not converge in the intrinsic valuation topology either.

For two set-sized hierarchies on two measurable spaces, take both exponent families in one ordered group Γ , and let f be a bounded real function measurable for the product sigma-algebra. Their well-ordered supports have a well-ordered sum and finitely many decompositions at each exponent by the Hahn product lemma. Thus the component product measures and monomial weights define a positive normalized law, with normalizer $Z_P Z_Q$, even for uncountable hierarchies. Coincident exponents combine by finite sums. Ordinary Fubini on each component product then proves the corresponding coefficientwise identity. Explicitly, the iterated integral is defined through component pairing as

$$\frac{1}{Z_P Z_Q} \sum_{\alpha, \beta}^H t^{\gamma_\alpha + \delta_\beta} \int \int f(x, y) d\nu_\beta(y) d\mu_\alpha(x). \quad (11.5)$$

Reversing the two ordinary integrals leaves each component unchanged; Hahn multiplication by the fixed reciprocal normalizers then preserves equality. The support condition is automatic for these two hierarchies in a common value group. One does not apply the real-observable definition unchanged to an arbitrary Hahn-valued inner integral. This construction does not give a Fubini theorem for all surreal-valued functions.

11.3 A genuine obstruction for arbitrary surreal-valued observables

A pointwise Hahn-valued function can have one well-ordered support and measurable real coefficients while naive coefficientwise integration fails positivity. The distinction between real-valued observables and arbitrary surreal-valued observables is therefore substantive.

Theorem 11.3 (An uncountable-support positivity failure). *There are an ordinary countably additive probability μ and a coefficientwise measurable Hahn-valued function X with a common well-ordered support such that $0 < X < 1$ everywhere, all coefficient functions are bounded and integrable, but integrating each coefficient separately gives a negative Hahn value.*

Proof. Let the sample space be the first uncountable ordinal ω_1 , with the countable-cocountable sigma-algebra. Give countable sets measure zero and cocountable sets measure one. This is countably additive: a disjoint sequence either has no cocountable member, in which case its union is countable, or has exactly one, with all remaining members countable. Two cocountable

sets cannot be disjoint on the uncountable space ω_1 . These cases give respectively total mass zero and one.

Take the additive subgroup $\Gamma \subset \mathbf{No}$ generated by the set of surreal ordinals $\xi \leq \omega_1$, with its inherited order; it is set-sized because finite integer combinations of a set form a set. Use the associated Hahn field, with $t^\gamma = \omega^{-\gamma}$ in the canonical surreal realization. Set

$$X(\xi) = t^\xi - t^{\omega_1} \quad (\xi < \omega_1). \quad (11.6)$$

Because $\xi < \omega_1$, the first term is larger than the second; and $t^\xi \leq 1$. Thus $0 < X(\xi) < 1$. The common exponent set $\{\xi : \xi < \omega_1\} \cup \{\omega_1\}$ is well ordered. At exponent $\alpha < \omega_1$ the coefficient function is the indicator of $\{\alpha\}$; at exponent ω_1 it is the constant -1 . All are measurable and bounded. Every singleton coefficient integrates to zero, while the last coefficient integrates to -1 . The formal coefficientwise integral is consequently $-t^{\omega_1} < 0$. $\square$

The example asserts coefficientwise measurability, not measurability for an unspecified topology on the Hahn field. It exhibits precisely why that weak measurable-function class cannot support a positive coefficientwise integration operation. It does not contradict ordinary Lebesgue integration, the finite-range construction, or (11.4) for real-valued functions.

There is a useful boundary to this failure. If a pointwise nonnegative Hahn-valued function has a *countable* common well-ordered support and each coefficient is integrable against an ordinary positive finite measure, its coefficientwise integral is nonnegative. To see this, write $X = \sum_{\gamma \in W} x_\gamma t^\gamma$ and suppose first that the integral has a least nonzero coefficient at η . For $\gamma < \eta$ in W , transfinite induction shows $x_\gamma = 0$ almost everywhere. Indeed, outside the countable union of the earlier exceptional null sets, all preceding coefficients vanish. Hahn positivity then forces $x_\gamma \geq 0$ there; its zero integral forces it to vanish almost everywhere. Apply the same argument at η : $x_\eta \geq 0$ almost everywhere and its nonzero integral is strictly positive. If there is no nonzero integral coefficient, the integral is zero. This proves the claim, including finite and empty supports. Applying it also to $M - X$ gives $0 \leq \int X d\mu \leq M\mu(\Omega)$ when $0 \leq X \leq M$; for a probability measure this preserves the bound M . The failure in (11.6) occurs at exponent ω_1 , where removing the earlier exceptional singletons would remove the whole sample space: an uncountable union of null sets need not be null.

11.4 Posterior kernels in a finite diffuse hierarchy

A conditional distribution for a continuous observation needs more than a ratio of event masses at a singleton. Here is a construction with its integration convention made explicit. Let $\mu_0, \dots, \mu_m$ be ordinary joint probability laws of (X, Y) on standard Borel spaces. Suppose their ordinary conditional kernels $K_j(y, A)$ and marginal measures μ_j^Y have been selected, so that

$$\mu_j(X \in A, Y \in B) = \int_B K_j(y, A) d\mu_j^Y(y). \quad (11.7)$$

The usual real disintegration theorem supplies this input in this setting [12, Theorem 8.5], applied with the observation as the conditioning variable and the state as the Borel-valued variable. It is an ordinary real disintegration input.

Take $\rho = (m+1)^{-1} \sum_j \mu_j^Y$ and let $h_j = d\mu_j^Y/d\rho$. Choose versions which are finite and nonnegative on a common full- ρ measurable set G . By uniqueness of Radon–Nikodym derivatives, $\sum_j h_j = m+1$ almost everywhere; shrink G to enforce this equality everywhere on G . Thus $0 \leq h_j \leq m+1$ there and at least one h_j is positive. Set every h_j to zero off G . For $y \in G$ and

$0 = \gamma_0 < \dots < \gamma_m$, put

$$D(y) = \sum_{j=0}^m t^{\gamma_j} h_j(y), \quad (11.8)$$

$$P_y(A) = \frac{\sum_{j=0}^m t^{\gamma_j} h_j(y) K_j(y, A)}{D(y)}. \quad (11.9)$$

For each $y \in G$, let j be the first index with $h_j(y) > 0$. Factoring $t^{\gamma_j} h_j(y)$ out of $D(y)$ leaves one plus a finite positive-exponent series. The denominator is therefore positive, and expansion of its inverse makes each coefficient of P_y a finite real linear combination of the ordinary probability measures $K_k(y, \cdot)$. This proves coefficientwise countable additivity in A ; positivity and normalization follow directly from the quotient. Its standard part is $K_j(y, \cdot)$.

There is also a common support bound as y varies. On the measurable piece with first active index j , the inverse expansion uses the additive monoid generated by the positive gaps $\gamma_k - \gamma_j$ for $k > j$. This monoid is well ordered with finitely many contributions to each coefficient by the Neumann lemma. The finite union over j is again well ordered. Every coefficient of $P_y(A)$ is measurable in y , being a finite sum of products and quotients of measurable functions with $h_j(y) > 0$ on that piece. Extend P_y by any fixed ordinary probability law off G ; this adds at most exponent zero and preserves coefficientwise measurability.

The precise reconstruction identity is

$$P(X \in A, Y \in B) = Z^{-1} \int_B D(y) P_y(A) d\rho(y) = Z^{-1} \sum_{j=0}^m t^{\gamma_j} \int_B h_j(y) K_j(y, A) d\rho(y), \quad (11.10)$$

where $Z = \sum_j t^{\gamma_j}$. In the middle expression the product $D(y)P_y(A)$ is formed *before* coefficientwise integration; it cancels the denominator, giving exactly the finite sum on the right. Those coefficient functions are integrable, since $0 \leq K_j \leq 1$ and h_j is a probability density. Coefficients of the quotient $P_y(A)$ alone need not have the integrability needed for an arbitrary termwise integration operation. For an explicit example, take $Y \in (0, 1)$, $X \in \{0, 1\}$, and two components: under μ_0 , $X = 0$ and Y has density $2y$; under μ_1 , $X = 1$ and Y has density $2(1 - y)$. Then ρ is Lebesgue probability, $h_0(y) = 2y$, $h_1(y) = 2(1 - y)$, and weights $1, t$ give

$$P_y(\{1\}) = \frac{t(1 - y)}{y + t(1 - y)}.$$

Its coefficient at t is $(1 - y)/y$, whose real integral diverges at zero. Yet $D(y)P_y(\{1\}) = 2t(1 - y)$ is integrable coefficientwise, and reconstruction gives $P(X = 1) = t/(1 + t)$. Boundedness of the posterior as a Hahn value does not bound its separate real coefficients.

Equation (11.10) is a rigorous posterior construction for this model, not a general surreal Radon–Nikodym or disintegration theorem. Its version choices on sets null for every component remain real choices. Replacing scalars does not determine a unique conditional law on observations which the entire model assigns exactly zero mass.

12 Two coherent coin models with no coefficientwise path law

For each ordinary finite n , independent coin weights are legitimate surreal probabilities. The following proofs show why this is not enough for the coefficientwise extension in Definition 10.3. They concern $t = \omega^{-1}$ and the ordinary countable path space $\{0, 1\}^{\mathbb{N}_{\geq 1}}$ with its Borel sigma-algebra. Summing over a new coordinate multiplies any earlier pattern weight by $p + (1 - p) = 1$, so the finite laws are consistent; every pattern weight is positive for both biases. The obstructions below concern a real coefficient measure's finite total variation. Positivity of the proposed Hahn

extension is not used, so signed extensions are excluded as well. The conclusions are special cases of the repository's broader product analysis [13]; Section 1.5, item (c), names the corollary of that report which contains both and the mechanisms the proofs share.

Theorem 12.1 (A repeated infinitesimal success probability). *The consistent finite-dimensional laws of independent Bernoulli(t) coordinates do not extend even to a coefficientwise Hahn signed measure with ordinary finite signed coefficient measures.*

Proof. For $n \geq 1$, let $B_n = \{x : x_1 = \cdots = x_{n-1} = 0, x_n = 1\}$. The initial zero string is empty at $n = 1$. These measurable cylinder events are pairwise disjoint: a path cannot have two different first success times. Their prescribed finite-dimensional laws require

$$P(B_n) = t(1 - t)^{n-1}. \quad (12.1)$$

The coefficient of t is one for every n . If a coefficientwise law existed, its exponent-one coefficient would be an ordinary finite signed measure μ_1 with $\mu_1(B_n) = 1$ for all n . For any positive integer N , its total variation would be at least $\sum_{n=1}^N |\mu_1(B_n)| = N$, contradicting finiteness. In fact the complementary event $C_N = \{x_1 = \cdots = x_N = 0\}$ has mass $(1 - t)^N$ and coefficient $-N$ at t . The finite partition $B_1, \dots, B_N, C_N$ therefore gives the sharper bound $|\mu_1|(\Omega) \geq 2N$. Given a proposed finite real total variation M , an ordinary integer $N > M/2$ is already a contradiction; no infinite sum of surreal event masses is taken. $\square$

The obstruction does not say that any finite coin experiment is inconsistent. Nor does it say that a regular finitely additive path law is impossible: Section 13 constructs such laws after field enlargement. It says that one specific coefficientwise rule cannot accommodate these finite-dimensional distributions.

Theorem 12.2 (A repeated infinitesimal perturbation of a fair coin). *The independent Bernoulli($1/2 + t$) finite-dimensional distributions also have no coefficientwise extension with finite signed coefficient measures.*

Proof. Fix an ordinary integer $n \geq 1$. A length- n pattern with k successes has probability $(1/2 + t)^k(1/2 - t)^{n-k}$. To obtain its coefficient at t , choose the linear term from exactly one factor: the k success factors contribute $+2^{-(n-1)}$ each, and the $n - k$ failure factors contribute $-2^{-(n-1)}$ each. Thus

$$2(2k - n)2^{-n}. \quad (12.2)$$

The total variation of the proposed exponent-one coefficient measure, evaluated on the length- n partition, must therefore be at least

$$\sum_{k=0}^n \binom{n}{k} |2(2k - n)2^{-n}| = 2\mathbb{E}|S_n|,$$

where $S_n = \varepsilon_1 + \cdots + \varepsilon_n$ under the ordinary uniform law on n independent signs $\varepsilon_i \in \{-1, 1\}$. There are $\binom{n}{k}$ patterns with $S_n = 2k - n$; this explains the partition multiplicities. Centering and independence make every product with an odd occurrence of some index have expectation zero. Consequently

$$\mathbb{E}S_n^2 = n, \quad \mathbb{E}S_n^4 = n + 6\binom{n}{2} = 3n^2 - 2n \leq 3n^2.$$

The fourth moment counts either one index four times or two indices twice each. Hölder with conjugate exponents $3/2$ and 3 , applied to $|S_n|^2 = |S_n|^{2/3}|S_n|^{4/3}$, gives

$$n \leq (\mathbb{E}|S_n|)^{2/3}(\mathbb{E}S_n^4)^{1/3}, \quad \mathbb{E}|S_n| \geq \sqrt{n/3}. \quad (12.3)$$

Cubing the first inequality gives $n^3 \leq (\mathbb{E}|S_n|)^2 \mathbb{E}S_n^4 \leq 3n^2(\mathbb{E}|S_n|)^2$; divide by $n^2 > 0$ and take the nonnegative real square root. A proposed finite variation M is contradicted by any ordinary integer $n > 3M^2/4$, since then $2\sqrt{n/3} > M$. Thus no finite signed coefficient measure has the required values on all cylinder partitions. The argument uses finite real moments, not a central limit theorem. $\square$

A logit parametrization does not remove this obstruction. If every independent coordinate has the common infinitesimal logit t , then $\sigma(t) = 1/2 + t/4 + O(t^3)$. Its exponent-one cylinder coefficients are one quarter of those in (12.2), and their variations are at least $\frac{1}{2}\sqrt{n/3}$, still unbounded. Likewise, put $u = \exp(-L)$ for positive infinite L . In canonical Hahn coordinates write its leading term as at^β , with $a > 0$ real and $\beta > 0$; u need not itself be a monomial. The shared success probability $\sigma(-L) = u/(1+u)$ has the same leading term, as does every first-success mass $u/(1+u)^n$. Its exponent- β coefficient is therefore the same positive real a for all n , again forcing unbounded coefficient variation. This argument uses the original Hahn coordinates and does not assume that a change of scalar variable preserves coefficientwise measure structure. Specifying logits does not supply a missing infinite-addition rule.

The unperturbed fair coin has an ordinary real product law, hence a coefficientwise law at exponent zero. An infinitesimal perturbation at every coordinate is not automatically a harmless perturbation of the infinite product: its first coefficient already has unbounded variation. This is a precise failure of extension, not a claim that every non-Archimedean Kolmogorov-type theorem fails. Existing theorems with other hypotheses and addition rules address different questions [6, 8, 13].

13 Regular all-subsets laws and extension inside the surreals

13.1 A relative ordered-field embedding lemma

To distinguish an obstruction in one workspace from absolute inconsistency, we need an explicit enlargement principle.

Lemma 13.1 (Set-sized relative field embedding). *Let $K \subseteq \mathbf{No}$ be a set-sized ordered subfield, and let L/K be a set-sized ordered field extension. There is an ordered-field embedding $j : L \rightarrow \mathbf{No}$ fixing K pointwise.*

Proof. Let M be an ordered real closure of L , still a set-sized field. The subfield K_0 of elements of M algebraic over K is a real closure of K : positive square roots and roots of odd-degree polynomials in M remain algebraic over K . The same argument in the real closed field $\mathbf{No}$ gives its real closure of K , consisting of the surreals algebraic over K . This is set-sized, since there are set many polynomials and each has finitely many roots. Uniqueness of ordered real closures identifies these two fields by an ordered isomorphism fixing K .

Choose and well order a transcendence basis of M over K_0 . Extend this initial embedding successively, keeping its domain a real closed subfield of M and its image real closed. At a step let F be the current domain and x the next basis element. This x is transcendental over F , because earlier real-closure steps added only algebraic elements over the field generated by earlier basis elements. The sets

$$\{j(a) : a \in F, a < x\}, \quad \{j(b) : b \in F, x < b\} \quad (13.1)$$

form a surreal cut: every left entry is smaller than every right entry. The cut property applies even if one side is empty. Take its simplest surreal y , which lies strictly between the two sides.

No element of $j(F)$ can equal y , since every element of F lies strictly on one side of x . A real closed ordered field has no proper ordered algebraic extension, so y is transcendental over $j(F)$.

Substitution $x \mapsto y$ therefore gives an injective field map on $F(x)$. It preserves order: a nonzero polynomial over a real closed field factors into a nonzero constant, linear factors, and monic irreducible quadratic factors. The latter are positive everywhere (by completing the square), the linear signs are determined by the cut, and the constant's sign is already preserved. Apply this to numerator and denominator to preserve signs of rational functions. The ordered embedding extends uniquely to the real closure of $F(x)$ inside M , with image its real closure inside **No**.

At a limit stage take unions. Any positive element or odd-degree polynomial has all its data in one earlier stage, where its required root already exists, so the union remains real closed. After the whole basis has been included, M is algebraic over the resulting real closed subfield and hence equals it. Restrict the resulting embedding of M to L . Every stage, image and cut is a set, since the basis is set-sized and algebraic closure steps adjoin set many elements. The original field K is fixed throughout. $\square$

The map preserves field operations and order. It need not preserve a previously chosen exponential on L , an internal-set structure, or a specified Hahn summation operation. After embedding we may take the canonical exponential closure from Lemma 2.1, but that is not a proof that two different exponential structures agree.

13.2 A fine-ultrafilter construction

The following is a direct version of the non-Archimedean sampling construction in the literature: compare the fine-ideal construction in [6, Section 4.2] and the fine-ultrafilter regularity and uniformity results in [8, Definition 4 and Propositions 5–6]. The sections and numbers refer to the cited arXiv preprints. Let Ω be an infinite set and let I be its nonempty finite subsets. For each finite $F \subseteq \Omega$, the cone $\{s \in I : F \subseteq s\}$ is nonempty, and finite intersections of cones are cones: the intersection for $F_1, \dots, F_r$ is the cone for their finite union. The empty set is in no such filter because every cone has a nonempty finite member. Extend this proper cone filter to an ultrafilter $\mathcal{U}$. It is fine, meaning that every cone containing one prescribed point belongs to it. It cannot be principal at any $s \in I$, since a point outside that finite s has a cone omitting s . This use of the ultrafilter lemma is nonconstructive.

The ultrapower $L = \mathbb{R}^I / \mathcal{U}$ is a set-sized ordered field; reals embed as constant functions. Its order is defined by $[f] < [g]$ exactly when $\{s : f(s) < g(s)\} \in \mathcal{U}$. In particular a positive class need not have a positive real lower bound. Define

$$P(A) = \left[\frac{|A \cap s|}{|s|} \right]_{\mathcal{U}}, \quad H = [|s|]_{\mathcal{U}}, \quad A \subseteq \Omega. \quad (13.2)$$

These values give an all-subsets regular finitely additive probability. Indeed normalization and disjoint additivity hold at every snapshot. For $a \in A$, every snapshot in its cone has $|A \cap s|/|s| \geq 1/|s| > 0$, so $P(A) > 0$ in the ultrapower order. For every ordinary n , choose $n+1$ distinct points of the infinite set Ω ; their cone forces $|s| > n$. On the cone containing a , the singleton's snapshot frequency is exactly $1/|s|$. Thus

$$H > n \quad (n \in \mathbb{N}), \quad P(\{a\}) = 1/H \quad (a \in \Omega). \quad (13.3)$$

Lemma 13.1, applied over $\mathbb{R}$, turns these into surreal probability values without changing any field identities or real constants. The reciprocal $1/H$ is positive infinitesimal, since H exceeds every ordinary integer and hence every real bound. This yields positive equal singleton masses even on an uncountable set. Every finite set F has mass $|F|/H$, by finite additivity.

The addition implicit in (13.2) is a snapshot operation:

$$P(A) = \left[\sum_{a \in A \cap s} \frac{1}{|s|} \right]_{\mathcal{U}}. \quad (13.4)$$

It is not a strong Hahn sum of the identical surreal numbers $1/H$. The finite terms being summed depend on the snapshot before passage to the quotient. Erasing this dependence would change the construction and produce the strong-summability obstruction of Section 10.

13.3 Nonuniqueness and limited symmetry

On $\mathbb{N}_{\geq 1}$ one can instead use the initial segments $s_n = \{1, \dots, n\}$ and a nonprincipal ultrafilter on the indices. This remains fine for every fixed finite subset of $\mathbb{N}$. If the ultrafilter concentrates on even indices, the even numbers have probability exactly $1/2$, because the even count in s_n is $n/2$ on those indices. Such nonprincipal ultrafilters exist by extending the cofinite filter together with the infinite set of even indices. Using odd indices instead, the count is $(n-1)/2$, giving probability

$$\frac{1}{2} - \frac{1}{2H}, \quad H = [n]. \quad (13.5)$$

For either model the ordered field $\mathbb{R}(H)$ can be identified with $\mathbb{R}(\omega)$ by $H \mapsto \omega$: both generators are positive infinite and transcendental over $\mathbb{R}$, so polynomial signs are determined by their leading terms. Lemma 13.1 then embeds the full field over that identification. The two resulting surreal probabilities have the same chosen singleton mass $1/\omega$ and the same real probability of the evens, but different exact probabilities of the evens. Thus even fixing the infinitesimal unit need not select a unique law.

Equal singleton probabilities imply invariance under permutations with finite support. If a permutation moves only points in a finite set F , it fixes $A \setminus F$ and preserves the cardinality of $A \cap F$; finite additivity therefore preserves $P(A)$. This does not give invariance under arbitrary transformations. On $\mathbb{N}_{\geq 1}$ the shift $n \mapsto n+1$ is an injection, not a permutation; invariance under that injection would give $P(\mathbb{N}_{\geq 1} \setminus \{1\}) = P(\mathbb{N}_{\geq 1})$, contradicting $P(\{1\}) > 0$. Even invariance under all permutations of $\mathbb{N}_{\geq 1}$ is impossible: partition it into two infinite sets A, C and choose $a \in A$. Bijections $A \rightarrow A \setminus \{a\}$ and $C \rightarrow C \cup \{a\}$ combine into a permutation, under which $P(A)$ decreases by $P(\{a\}) > 0$. More generally, an invariance axiom can conflict with regularity even when every atom has the same mass. Symmetry requirements must be checked as separate constraints, not inferred from the word “uniform.”

13.4 Regular extension of any regular event algebra

Theorem 13.2 (All-subsets regular extension after field enlargement). *Let Ω be a nonempty set, $\mathcal{A}$ an algebra of its subsets, and $P_0 : \mathcal{A} \rightarrow K$ a regular finitely additive probability, where $K \subseteq \mathbf{No}$ is a set-sized ordered subfield. There is a set-sized ordered field extension L/K and a regular finitely additive probability $P : \mathcal{P}(\Omega) \rightarrow L$ extending P_0 . The extension can be realized with values in a set-sized subfield of $\mathbf{No}$ while preserving every original value in K .*

Proof. Use the first-order language of ordered fields with constants for the elements of K and a constant p_A for each subset $A \subseteq \Omega$. This is a set-sized language because K and $\mathcal{P}(\Omega)$ are sets. Include the ordered-field axioms and the atomic diagram of K with inequalities and distinctions between different constants, so that their interpretations form an ordered copy of K . Add $p_{\emptyset} = 0$, $p_{\Omega} = 1$, every disjoint finite-additivity identity, $p_A > 0$ for every nonempty A , and $p_A = P_0(A)$ for $A \in \mathcal{A}$. This is a set of first-order sentences.

Every finite fragment is satisfiable in K . To show this, take the finite Boolean algebra generated by the finitely many original events from $\mathcal{A}$ named in the fragment. Each of its nonempty atoms has positive P_0 mass. Refine its atoms by all the other finitely many subsets mentioned in the fragment, including every union appearing in an additivity identity. Within a parent atom C of mass $b = P_0(C) > 0$, let $r_C \geq 1$ be the number of its nonempty refined children and give each child mass b/r_C . This is positive in K , and their total is b . Assign each mentioned event the sum of the masses of its children. Every nonempty mentioned event contains a nonempty child, so its assigned mass is positive. Each original event is a union of parent atoms and therefore retains its P_0 mass. Disjoint unions correspond to disjoint lists of children, satisfying every additivity constraint in the fragment; the whole space has mass one and the empty set zero. Constants for unmentioned events may be interpreted arbitrarily for this fragment. The constants from K retain their given values. The assignments for different finite fragments need not agree; compactness is what produces one simultaneous model.

First-order compactness gives a set-sized ordered-field model of all these axioms. Its copy of K is an ordered subfield, and the values of the p_A define the desired extension. Restricting to the subfield generated by those values and K is enough. Finally apply Lemma 13.1 over K ; its image remains a set-sized field and all original probabilities are fixed. $\square$

The logical compactness theorem is the standard first-order theorem; no topological compactness of a surreal simplex is being assumed. Regularity is necessary if a regular extension must preserve every original mass: an original nonempty zero-mass event cannot become positive. The theorem does not require countable additivity, symmetry, a canonical embedding, or a prescribed bound on the new scales. Imposing such extra conditions may block the construction. If Ω is finite, the same splitting can be done once down to singletons, giving an extension already in K ; the compactness argument is needed for the unrestricted infinite case. Regularity is an actual hypothesis on the original event algebra, not a property supplied by changing scalar fields.

Corollary 13.3 (The two coin models do have regular finite-additive laws). *The finite cylinder probabilities of either $\text{Bernoulli}(t)$ or $\text{Bernoulli}(1/2 + t)$ extend to regular finitely additive surreal probabilities on all subsets of $\{0, 1\}^{\mathbb{N}_{\geq 1}}$, with the original $t = \omega^{-1}$ unchanged.*

Proof. Every cylinder-algebra event depends on finitely many coordinates and is a union of full patterns on those coordinates. For either bias, $0 < p < 1$, so each pattern has positive mass $p^k(1-p)^{n-k}$. Refining to any larger finite coordinate set preserves masses by $p + (1-p) = 1$, proving that this assignment is well defined and finitely additive. A nonempty event contains a pattern and therefore has positive mass. All these finite probabilities lie in $K = \mathbb{R}(t) \subseteq \mathbf{No}$. Apply Theorem 13.2 over this K and use the embedding that fixes K pointwise, in particular the original t . $\square$

Theorems 12.1 and 12.2 prove that these extensions cannot carry the specified coefficientwise finite-measure structure. There is no contradiction: the extension theorem promised a different axiom system. It also does not compute the probability of an arbitrary tail event. Such probabilities can depend on which model of the extension axioms has been selected.

14 An infinite learning model with a rare latent regime

The previous obstructions should not leave the impression that no interesting infinite stochastic models survive coefficientwise addition. Mixtures of ordinary path laws provide explicit counterexamples to that impression, and also expose a limitation of finite Bayesian updating.

Index coordinates by the positive integers. Let μ_0 be the ordinary independent Bernoulli(1/3) law on $\{0, 1\}^{\mathbb{N}_{\geq 1}}$ and μ_1 the independent Bernoulli(2/3) law, both on the product sigma-algebra. Attach a latent label $H \in \{0, 1\}$ and give the joint space the Hahn law

$$P = \frac{\delta_0 \otimes \mu_0 + t\delta_1 \otimes \mu_1}{1+t}. \quad (14.1)$$

Theorem 11.1 makes this a positive coefficientwise probability on its product sigma-algebra. It has an exact rare regime $P(H = 1) = t/(1+t) > 0$, not merely a family of finite-dimensional proposals. Its unconditional coordinate means and covariances are

$$\mathbb{E}X_i = \frac{1+2t}{3(1+t)}, \quad \text{Cov}(X_i, X_j) = \frac{t}{9(1+t)^2} > 0 \quad (i \neq j). \quad (14.2)$$

Indeed, independence within each component gives, for $i \neq j$,

$$\mathbb{E}(X_i X_j) = \frac{1+4t}{9(1+t)}, \quad \text{Cov}(X_i, X_j) = \frac{(1+4t)(1+t) - (1+2t)^2}{9(1+t)^2} = \frac{t}{9(1+t)^2}.$$

Each component is invariant under finite coordinate permutations, so the mixture is exchangeable. Its positive covariance rules out independence. Therefore its infinitesimally shifted marginals do not contradict Theorem 12.2 or the analogous coefficientwise independent-product obstructions.

Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ for ordinary integers $n \geq 0$, with $\mathcal{F}_0$ trivial, and write $K_n = \sum_{i=1}^n X_i$ (so $K_0 = 0$). An observation word of length n with k successes has likelihoods $(1/3)^k (2/3)^{n-k}$ and $(2/3)^k (1/3)^{n-k}$ in the two regimes. Both are positive, and their ratio is 2^{2k-n} . Every atom of $\mathcal{F}_n$ therefore has positive probability. Bayes' rule gives the finite conditional probability of the rare regime:

$$M_n = P(H = 1 \mid \mathcal{F}_n) = \frac{t 2^{2K_n-n}}{1+t 2^{2K_n-n}}. \quad (14.3)$$

Every value of M_n is positive infinitesimal for each ordinary n : the factor $r_n = 2^{2K_n-n}$ is a positive real number. In particular, $M_0 = t/(1+t)$ and $0 < M_n < 1$. These adapted finite-range variables form a bounded martingale by the finite tower property. One can also check this directly. Given a word with likelihood ratio r , the next success probability and the two next posterior values are

$$q = \frac{1+M_n}{3} = \frac{1+2tr}{3(1+tr)}, \quad M_{n+1}^{(1)} = \frac{2tr}{1+2tr}, \quad M_{n+1}^{(0)} = \frac{tr}{2+tr}.$$

Then $qM_{n+1}^{(1)} + (1-q)M_{n+1}^{(0)} = tr/(1+tr) = M_n$. All these conditional expectations use finite partitions; no infinite surreal sum or martingale convergence theorem is used.

The ordinary strong laws of large numbers of the two real component measures [12] give the Borel event

$$B = \left\{ \lim_{n \rightarrow \infty} \frac{K_n}{n} = \frac{2}{3} \right\}, \quad \mu_0(B) = 0, \quad \mu_1(B) = 1. \quad (14.4)$$

Here an elementary real argument suffices for the imported strong law. For $h \in \{0, 1\}$, write $\mathbb{E}_h$ for expectation under μ_h , and put $p = (1+h)/3$, $Y_i = X_i - p$. Independence, centering and $\mathbb{E}_h Y_i^2 \leq 1/4$, $\mathbb{E}_h Y_i^4 \leq 1$ give, for $n \geq 1$,

$$\mathbb{E}_h \left(\sum_{i=1}^n Y_i \right)^4 = \sum_{i=1}^n \mathbb{E}_h Y_i^4 + 6 \sum_{i < j} \mathbb{E}_h Y_i^2 \mathbb{E}_h Y_j^2 \leq n + \frac{3n(n-1)}{16} \leq n^2.$$

For each real $\varepsilon > 0$, Markov's inequality therefore bounds the component probability of $|K_n/n - p| \geq \varepsilon$ by $1/(n^2 \varepsilon^4)$. The probability of a violation at some $n \geq N$ is bounded by the

tail of this convergent real series, which tends to zero. Thus violations occur only finitely often almost surely. Intersecting these full-measure events for $\varepsilon = 1, 1/2, 1/3, \dots$ proves $K_n/n \rightarrow p$. This argument is entirely within the two real component measures.

The convergence condition defining B uses countably many inequalities between measurable real functions, so B is measurable. For every fixed m , the same limit is obtained from $\sum_{i=m+1}^n X_i/n$, which depends only on coordinates after m . Hence B belongs to every future-coordinate sigma-algebra and is a tail event. We also write B for its inverse image in the joint space. Consequently

$$P(B) = \frac{t}{1+t} > 0, \quad P(H = 1 \mid B) = 1. \quad (14.5)$$

Proposition 14.1 (Finite-update shadows need not recover rare tail conditioning). *In model (14.1), every finite posterior in (14.3) has standard part zero, but observing the tail event B gives conditional probability one to $H = 1$. On B , the finite posteriors do not converge to that conditional value even in the intrinsic ordered-field topology.*

Proof. The finite-posterior assertion was proved above. The component strong laws of large numbers give (14.5). For every ordinary n and every path, $M_n < 1/2$ because it is infinitesimal. Thus on B its distance from the value one remains greater than $1/2$. $\square$

This also has a precise conditional-expectation formulation. Put $\mathcal{F}_\infty = \sigma(X_1, X_2, \dots)$. For every observation event A ,

$$P(H = 1, A) = \frac{t\mu_1(A)}{1+t} = \frac{\mu_0(B \cap A) + t\mu_1(B \cap A)}{1+t} = P(B \cap A),$$

because $\mu_0(B) = 0$ and $\mu_1(B^c) = 0$. Thus the $\mathcal{F}_\infty$ -measurable variable $\mathbf{1}_B$ satisfies $\mathbb{E}[\mathbf{1}_A \mathbf{1}_{\{H=1\}}] = \mathbb{E}[\mathbf{1}_A \mathbf{1}_B]$ for every $A \in \mathcal{F}_\infty$. This is precisely the defining integral identity for a version of conditional expectation, using only finite-range integrals. Also $P(\{H = 1\} \triangle B) = 0$. The martingale (M_n) does not recover this version on B even at a fixed real resolution.

Off B , convergence to zero fails at infinitesimal resolution as well. In the fixed chart $t = \omega^{-1}$, each M_n has valuation one and leading coefficient $r_n > 0$, so $M_n > t^2$ at every ordinary time and on every path. More explicitly, if the next observation is 1 or 0, respectively, then

$$M_{n+1} - M_n = \begin{cases} \frac{tr_n}{(1+2tr_n)(1+tr_n)}, & X_{n+1} = 1, \\ -\frac{tr_n}{(2+tr_n)(1+tr_n)}, & X_{n+1} = 0. \end{cases}$$

The leading coefficient of the absolute difference is r_n or $r_n/2$, respectively, and its valuation is one. Consequently $|M_{n+1} - M_n| > t^2$ everywhere. The single order tolerance t^2 therefore violates the Cauchy condition on every path, regardless of how small or large the real factors r_n become. These are ordinary integer times; no transfinite extension of the process is being asserted.

By contrast, the posterior standard parts are identically zero. Their ordinary limit differs from $\mathbf{1}_B$ exactly on B , whose probability is positive infinitesimal but whose real shadow probability is zero. This is fully consistent with the ordinary real martingale convergence theorem. A surreal analogue retaining all positive infinitesimal events or all infinitesimal resolutions must specify a different conclusion or impose additional hypotheses.

Coefficientwise convergence gives a third, different answer. At each ordinary time the geometric expansion is

$$M_n = \sum_{j \geq 1}^H (-1)^{j-1} r_n^j t^j.$$

On the event $B_0 = \{K_n/n \rightarrow 1/3\}$, which has μ_0 -measure one, $(2K_n - n)/n \rightarrow -1/3$, so $r_n \rightarrow 0$ in $\mathbb{R}$. Every fixed coefficient then tends to zero: M_n converges coefficientwise to zero on B_0 , although it is nowhere order-Cauchy. On B , instead, $(2K_n - n)/n \rightarrow 1/3$ and $r_n \rightarrow +\infty$; already the coefficient of t has no finite real limit. This coefficientwise statement is a limit in $\mathbb{R}$ at each fixed exponent, not strong Hahn summation of the time sequence.

The example also distinguishes two uses of “infinitely much evidence.” A Borel tail observation is an event in a specified path law. It is not an unrestricted surreal-indexed sum of finite log-likelihood increments. The first has a rigorous interpretation here; the second has not been defined merely by writing an infinite logit.

15 Standard-part bridges, Loeb measure, and sampling limits

15.1 Exactly when the real shadow is countably additive

Theorem 15.1 (A continuity test for the shadow). *Let P be a finitely additive K -probability on a sigma-algebra. Then $P_0 = \text{st } P$ is countably additive exactly when, for every $A_n \downarrow \emptyset$ and every positive real ε , there is an n with $P(A_n) < \varepsilon$.*

Proof. A real finitely additive probability is countably additive exactly when it is continuous from above at the empty set. For completeness, countable additivity gives this continuity by decomposing A_1 into successive differences. Conversely, for disjoint B_n with union B , the remainder $A_N = B \setminus \bigcup_{n \leq N} B_n$ decreases to the empty set, so finite additivity and continuity give $P_0(B) = \sum_n P_0(B_n)$.

If $P_0(A_n) \rightarrow 0$, choose n with $P_0(A_n) < \varepsilon/2$. The infinitesimal difference between $P(A_n)$ and its standard part is less than $\varepsilon/2$, so $P(A_n) < \varepsilon$. Conversely, the displayed condition implies that the decreasing real numbers $P_0(A_n)$ have infimum at most every positive real ε , hence zero. This is the required continuity. $\square$

The standard part of the normalized hierarchy is the ordinary measure μ_0 and satisfies this test. The all-subsets uniform law on $\mathbb{N}$ from (13.2) does not: singleton shadows are zero while the shadow of their union is one. More explicitly, for $A_n = \{n+1, n+2, \dots\}$, one has $P(A_n) = 1 - n/H$ and $P_0(A_n) = 1$. Standard part is always a finite-additive homomorphism of probability values; countable additivity is an extra property, not an automatic consequence of the existence of standard part.

15.2 The Loeb bridge requires internal structure

Loeb’s construction starts with an internal finitely additive probability on an internal algebra in a sufficiently saturated nonstandard setting [11]. Its limited values have real standard parts. On the internal algebra these define a real premeasure, which extends to a countably additive measure on the generated sigma-algebra and then to its completion. This real completed measure is the Loeb measure.

The role of saturation can be seen directly in the finite-probability case. For a decreasing sequence of internal sets with empty intersection, countable saturation forces some member to be empty: otherwise their finite intersections would all be nonempty. In particular, if countably many disjoint internal sets have an internal union, only finitely many can be needed to cover that union. Standard-part finite additivity therefore supplies the premeasure property on the internal algebra. The ordinary measure-extension theorem then supplies the sigma-algebra and its countable additivity. These steps concern internal *sets* as well as non-Archimedean numbers.

An ordered-field embedding of the underlying scalar field into **No** preserves standard parts because it fixes $\mathbb{R}$ and preserves order. Thus one may carry an already specified internal model and its Loeb construction along with such a scalar embedding. The embedding alone, however, does not manufacture internal events, saturation, hyperfinite sets, or transferred sums. A bare surreal $H > \mathbb{N}$ is not a substitute for this package of structure.

15.3 A Poisson shadow at reciprocal-probability sample size

Here is a concrete sampling limit with all semantics supplied. Fix a positive real λ . For each ordinary integer $N > \lambda$, perform N independent Bernoulli trials with real success probability $p_N = \lambda/N$. Let S_N be the count and pass along a nonprincipal ultrafilter on these indices to the corresponding internal experiments. Write $H = [N]$ for the hyperinteger sample size. A single trial then has probability λ/H .

For a fixed ordinary integer $k \geq 0$, the internal event that the count is k has probability represented by

$$b_k = \left[\binom{N}{k} \left(\frac{\lambda}{N} \right)^k \left(1 - \frac{\lambda}{N} \right)^{N-k} \right], \quad \text{st } b_k = e^{-\lambda} \frac{\lambda^k}{k!}. \quad (15.1)$$

For $N < k$ one assigns the probability zero, which affects only finitely many indices. To verify the standard part, factor the binomial term as

$$\frac{\lambda^k}{k!} \prod_{r=0}^{k-1} \left(1 - \frac{r}{N} \right) \left(1 - \frac{\lambda}{N} \right)^{N-k}. \quad (15.2)$$

The finite product tends to one in ordinary real analysis. The logarithm of the last factor is $(N - k)(-\lambda/N + O(N^{-2})) \rightarrow -\lambda$. This proves the ordinary real limit, and every nonprincipal ultrapower of a convergent bounded real sequence has that standard part.

In a saturated internal realization, the Loeb probability of the count being an ordinary finite integer is one: the disjoint events with fixed counts k have the real Poisson masses in (15.1), whose ordinary sum is one. Thus the Poisson limit is recovered at a sample size comparable to the reciprocal of an infinitesimal success probability. This supplies the missing meaning of the informal scale calculation $n \asymp 1/p$ in a *chosen nonstandard experiment*.

After an ordered-field embedding into **No**, the probabilities and standard parts in (15.1) remain valid. But an internal power such as $[(1 - \lambda/N)^N]$ must not be silently identified with the canonical surreal expression $\exp(H \log(1 - \lambda/H))$ under a field-only embedding. The latter uses extra exponential structure. Equality requires an exponential-compatible representation, not just preservation of field operations. The fixed- k probability limits above do not require that extra claim.

16 An implementation and formalization architecture

16.1 Choose a representable scalar fragment

General surreal numbers are not finite data structures, and equality, leading-term extraction, or arbitrary infinite-support manipulation is not provided by a formal normal-form notation alone. A practical engine should start with a fragment such as rational functions in a positive infinitesimal, finite multiscale rational expressions, or explicitly supported Hahn series with a certified leading term. Canonical logarithms and exponentials may require a larger expression language than the probability fragment.

For a finite normalized model, a useful representation stores positive unnormalized weights rather than rounded probabilities. Normalization, conditioning, and Bayes then use exact

sums and quotients. The minimum-plus and leading-coefficient layer from Theorem 5.1 can be computed first, while residual expressions are retained for ties and later rare conditioning. A leading-scale answer should not be presented as an exact full probability.

Theorem 5.3 supplies a sufficient precision contract: when conditioning on mass of valuation β and requesting output through scale λ , require input error beyond $\beta + \lambda$. This uniform guarantee is not a necessary condition for each query; specific numerator–denominator cancellations can permit weaker precision. An implementation which first truncates all atoms at one fixed absolute threshold cannot guarantee every subsequent conditional query. For score-based models, Proposition 6.3 gives a different contract: an additive score error bound controls multiplicative probability error. This can be preferable when tiny probabilities must remain distinguishable.

16.2 Separate proof interfaces

A formal development should not require the full construction of **No** for every finite probability theorem. The following layers isolate the actual assumptions:

1. An ordered-field finite-probability layer: weights, finite sums, conditioning, expectation, inequalities, finite products, and stopping.
2. A residue/valuation layer: standard part, leading coefficients, conditional shadows, minimum-plus updates, and precision theorems.
3. An exponential ordered-field layer: logit, softmax, entropy, relative entropy, proper scoring, and the explicit Gibbs identity.
4. Separate infinite modules for strong supports, coefficient measures, compactness/ultrafilter extensions, and internal/Loeb constructions.

The first three modules should not import an unproved global integration operator. An interface for infinite sums should name its support or convergence certificate and state which regroupings it permits. A field embedding interface should distinguish preservation of order, exponential, standard part, strong sums, and internal structure. An ordered embedding fixing $\mathbb{R}$ preserves standard parts: the inequalities $r - \varepsilon < x < r + \varepsilon$ for every positive real ε are preserved. Preservation of exponentials, strong sums or internal structure requires additional hypotheses.

The repository has a maintained formalization ledger, but no new Lean proofs are delivered here. The present package’s executable tests are exact checks in a small rational-function fragment. They are evidence about those examples and the implementation, not a formal verification of the general mathematical theorems.

16.3 What the accompanying verifier checks

The file `code/verify.py` implements exact arithmetic in $\mathbb{Q}(t)$ with t ordered as a positive infinitesimal. It checks finite Bayesian normalization and update order, leading-scale posterior formulas, conditional shadows, conditional precision examples, finite probability inequalities, the rare-regime covariance, the hidden-shadow examples, and the explicit cylinder coefficients used in the nonextension proofs. The delivered run passed 2,145 assertions under Python 3.13.5, with the breakdown recorded in the JSON output. All comparisons use rational polynomial arithmetic, not substitution of a small floating-point number for t .

The output `data/verification.json` records the actual assertion count and test families. The program cannot check first-order compactness, uncountable sets, saturation, canonical surreal exponentiation, or an infinite measure-extension theorem. Those parts are justified by

the written proofs and explicitly cited foundational inputs, not by finite experiments. The separation is important: thousands of finite successful checks would not establish an infinite extension axiom.

On placement the unchanged program was rerun on a copy under Python 3.14.4. It passed the same 2,145 assertions in the same 19 families, and its output, written to a separate file, matched `data/verification.json` in every field except the recorded Python version.

17 Scope, limitations, and next mathematical questions

17.1 What has been built

There is a usable finite probability and decision theory with genuinely surreal-valued probabilities. It includes exact conditioning on positive infinitesimal events, canonical log-odds, finite information inequalities, and explicit optimization identities. It retains a hierarchy of conditional beliefs invisible to the unconditional real shadow. There are also genuine infinite-space models: strong atomic examples, diffuse coefficientwise hierarchies, a rare latent path regime, and regular all-subsets finite-additive extensions. The relationships among these constructions are mathematical statements with different hypotheses, not an assertion that one is the uniquely correct probability theory.

Several results identify useful boundaries rather than failures of the whole program. The conditioning precision theorem explains which approximations are safe. The coin obstructions identify coefficient variation as an extension barrier. The uncountable-support example shows that coefficientwise measurability alone does not define a positive integration theory. The latent-regime example separates finite posterior shadows from conditioning on a tail observation. These are concrete requirements for any stronger theory.

17.2 What is not asserted

We have not constructed a countably additive probability on all events using the full fine topology of **No**; Proposition 10.1 explains why ordinary set-indexed convergence is unsuitable there. We have not proved a universal Carathéodory, Radon–Nikodym, Fubini, martingale convergence, strong law of large numbers, central limit, or stochastic calculus theorem after unrestricted replacement of $\mathbb{R}$ by **No**. Each such theorem needs a specified infinite framework. The Poisson and latent examples import ordinary component laws or a stated nonstandard model; they are not substitutes for those missing general hypotheses.

No theorem here assigns a unique answer to every exact-zero conditioning problem, selects a canonical ultrafilter, or reconciles regularity with all possible symmetry demands. No statement makes arbitrary surreal expressions effectively computable. No assertion identifies the Conway omega-map with arbitrary exponential powers or transfers internal hyperfinite operations through a field-only embedding. The information and Gibbs results concern an ordinary finite alphabet, not an unspecified sum over a proper class or infinitely many states.

17.3 A concrete research agenda

One productive question is to characterize positive integration domains for coefficientwise Hahn probabilities beyond bounded real observables and countable-support functions. Theorem 11.3 shows that a common well-ordered support and integrable coefficients are insufficient at uncountable rank. A stronger condition could require simultaneous almost-everywhere control along support filtrations, with a precise closure theorem under products and conditioning.

A second question is to formulate convergence of conditional beliefs that preserves selected infinitesimal regimes without demanding full fine convergence. The rare latent example is a useful test: a proposed martingale theorem must explain whether its conclusion is taken modulo standard-null sets, exact-null sets, coefficient ideals, or a chosen scale cutoff. These choices yield different, potentially useful notions of learning.

A third question concerns extension with additional structure. Theorem 13.2 guarantees regular all-subsets extension after field enlargement, but not invariance, a preferred support group, or an exponential-compatible embedding. Identifying exactly which finite constraints on these extra structures have simultaneous realizations would turn a broad existence theorem into a more informative modeling tool. The symmetry counterexample shows that the answer cannot simply be “all natural constraints.”

Finally, a verified computational library can make substantial progress without solving every foundational problem. The finite ordered-field core, valuation precision contracts, logit identities, and finite information inequalities have explicit proofs and limited interfaces. They form a tractable starting point for formalization and exact symbolic experiments while the infinite theories remain deliberately separate.

17.4 Conclusion

Surreal probability is most useful as a multiscale extension of ordinary probability, not as a license to ignore the meaning of infinite addition. Its finite algebra is robust: positive infinitesimal events can be conditioned upon, infinite logits retain nonzero alternatives, and information and optimization identities remain exact. Its infinite analysis is plural: support, coefficient variation, internal structure, and standard-part continuity decide which familiar theorems survive. Once those choices are explicit, the resulting theory is mathematically substantial and operationally precise, while its limitations become statements that can themselves be proved and tested.

A Notation and dependency ledger

Symbol	Meaning
K	Set-sized ordered subfield of No containing $\mathbb{R}$
E	Such a field closed under canonical exponential and positive logarithm
t^γ	Hahn monomial identified with Conway $\omega^{-\gamma}$
$v(x), \text{lc}(x)$	Least support exponent and its coefficient, for $x \neq 0$
$\text{st } x$	Real standard part of a limited surreal
$\sum^H$	Strong Hahn sum; not an ordinary convergent real series
P_0	Unconditional real shadow $\text{st } P$
$C(A \mid B)$	Real shadow of the exact conditional ratio
λ_γ	$-\log(t^\gamma)$, with canonical logarithm
σ, logit	Logistic map and its inverse on $(0, 1)$
$D(p \parallel q)$	Finite relative entropy, under the support condition
H in ultraproducts	A specified internal sample-size class, not an arbitrary ordinal

The algebraic results in Sections 3–5 need only ordered fields plus the explicitly used residue or Hahn structure. Sections 6–8 additionally use canonical exponential and its scalar inequalities. Section 11 uses ordinary real finite measures and Hahn support multiplication. Section 13 uses choice, ordered real closures, the surreal cut property, and first-order compactness. Section 14 imports two ordinary real product laws and their strong laws of large numbers. Section 15 distinguishes ordinary real measure extension from the extra saturation assumption in the

Loeb construction. No infinite result is deduced solely from the finite exponential-field transfer principle.

B Reproduction and evidence boundaries

The source is standalone and uses an internal bibliography, with no external images, bibliography processor, or downloaded fonts. In the directory of this report, build and check with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python3 code/verify.py --output rerun-verification.json
```

When `latexmk` is unavailable, run `pdflatex` repeatedly until cross-references and the table of contents stabilize. The program writes a file only when `-output` is given; naming a new file keeps the delivered record `data/verification.json` intact. The delivered build script `code/build.sh` is kept unchanged. As placed it does not work: it changes into its own directory `code/` and builds `surreal_probability.tex`, which is not there. The delivery was checked for unresolved references and typesetting errors, and the PDF was rendered for visual inspection; `data/BUILD_VALIDATION.json` records that check of the delivered 35-page PDF, before placement. A clean typesetting log does not certify mathematical correctness.

The research audit lists the consulted repository snapshot and primary sources. The repository is an evolving collection of research drafts; this article’s references identify the exact comparison point. The article’s own status is likewise a proof-bearing research exposition, not a claim of independent peer review or exhaustive literature priority. The finite verifier and the written infinite proofs provide different kinds of evidence and should not be conflated.

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Placement note. The three entries above that carry the manuscript’s pin are its own references. This report now belongs to the same collection, and Section 1.5 compares it with those reports as committed at 50cb709.